PREPRINT

One Field, Five Readings: Metaballs, Haar-Wavelet Texture, and the Color Grammar of a Generative Creature

César Galindo[a]

[a]Facultad de Empresariales, Universidad Panamericana, Ciudad de México, Mexico

ARTICLE HISTORY
Compiled October 25, 2026

ABSTRACT
A single scalar field on the plane, built from a symmetry-broken polar contour, a sigmoid interior, a sum of metaball wells and a Haar-wavelet texture, is turned by five color maps into five creatures named after Studio Ghibli figures (Totoro, Shishigami, Kodama, No-Face, Catbus). The shape is geometry; the reading is supplied by the palette. We make the geometry exact. The contour $1 + a\sin k\theta$ is convex exactly when $a \le 1/(k^2+1)$, so every lobe of the five creatures is a genuine concavity, bounded by $2k$ inflections, and the contour always has at least $2k$ vertices. Bilateral symmetry is a parity condition on the harmonic orders, and for an even order no phase restores it. Two metaball wells fuse along a Thom cusp: with weights α_1, α_2 at distance $2a$, the field has two maxima exactly when $a^2 > \varepsilon/3$ and $|\log(\alpha_1/\alpha_2)|$ is below an explicit threshold that vanishes like $(a^2 - \varepsilon/3)^{3/2}$, so that the birth of a second body from one is a cusp catastrophe. Level sets of the field change topology only at critical values, and the roughness exponent H of the Haar texture fixes how its variance is shared among octaves. On the color side, CIELAB analysis of the five palettes shows that chroma, hue spread and lightness range are all needed to separate them, that hue statistics must be computed only over colors with a hue, because hue is a polar coordinate singular at the achromatic axis, and that the single-hue palettes survive red–green color deficiency while the two multi-hue ones lose about two thirds of their contrast. The claim that needs viewers is stated as a falsifiable prediction, with the sample size needed to refute it.

KEYWORDS
generative geometry; metaballs; convexity; four-vertex theorem; cusp catastrophe; Haar wavelets; CIELAB; color vision deficiency

1. Introduction

Mathematics has used color as a teaching instrument for a long time. Byrne's 1847 Euclid replaced the lettered vertices of every diagram with colored angles, lines and figures "for the greater ease of learners" (Byrne 1847). Papert's turtle geometry made a drawing the object of study, so that a learner could change a number and watch a shape change (Papert 1980; Abelson and diSessa 1981). Sinclair's work on the aesthetic in mathematical inquiry adds that visual appeal motivates, evaluates and generates mathematical ideas, in children as in mathematicians (Sinclair 2004). What these traditions lack is a small object in which the mathematics of a picture and the color of the picture can be varied independently, so that a class can *see* which of the two is

CONTACT César Galindo. Email: jcgalindo@up.edu.mx

doing which job. This article proposes one.

The object is a scalar field $\Phi : \mathbb{R}^2 \to \mathbb{R}$ evaluated on a grid and mapped to color. Its mathematical content — symmetry, level sets, gradient, scale structure — is independent of the palette applied to it. That independence is usually a technical footnote; here it is the lesson. The field is assembled from four classical generative-art primitives: a symmetry-broken polar contour, a sigmoid-thresholded interior, a sum of metaball wells (Blinn 1982), and a multiresolution texture built from the Haar wavelet (Haar 1910) in the spirit of wavelet noise (Cook and DeRose 2005). Five parameter-and-palette choices are read by a viewer as five Studio Ghibli creatures: the gentle forest spirit Totoro, the ambivalent forest god Shishigami, the pale, near-still kodama tree spirits, the void-bodied, single-accented Kaonashi ("No-Face"), and the warm, wide Catbus.

The color half of the lesson has an empirical literature, and that literature is uneven. A systematic review of 132 studies (128 years, 42,266 participants, 64 countries) finds reliable, cross-cultural color–emotion associations along valence, arousal and power (Jonauskaite and Mohr 2025). An influential claim that seeing red impairs intellectual performance (Elliot et al. 2007) has not held up in replications and meta-analysis (Steele 2014; Gnambs 2020). A teaching design should rest on the first kind of result and not the second, so Section 2 grades the evidence before the article uses any of it.

The question this article puts to its own construction is narrow: given one fixed shape, how much of its emotional reading can color alone supply, and what should a teacher check before choosing a palette for a mathematical figure? Section 3 gives the field, Section 4 the exact geometry of its contour and of the fusion of its wells, and Section 5 its wavelet texture, each with a short *In the classroom* paragraph and a figure a student can regenerate by changing one parameter. Section 6 gives the five registers. Section 7 is the quantitative core: CIELAB statistics of the five palettes, a test of how legible each palette is to readers with red–green color-vision deficiency, and a ranking of the palettes by the Pleasure–Arousal–Dominance model of color. Section 8 maps every construction step to curriculum topics and states the one claim that needs students to test, as a falsifiable prediction.

2. Color psychology for a mathematics classroom: what the evidence supports

We searched the color–emotion and color–learning literature in September 2026 and opened the record of every source cited below. Two publisher pages that host eye-tracking studies of color cueing in mathematics problem solving refused automated access; those studies are therefore not cited. In what we could open, we found no classroom trial that varies the *affective* quality of a palette while holding the mathematics fixed. The article's design targets that gap and does not fill it. Table 1 is the ledger; the paragraphs below give the reasoning.

What is solid. Color–emotion associations replicate across countries, with linguistic and geographic proximity shaping the residual differences (Jonauskaite et al. 2020; Jonauskaite and Mohr 2025), and a plausible mechanism exists: preference tracks the affect of the objects a color evokes (Palmer and Schloss 2010). Brightness and saturation showed strong, consistent effects on emotion in the classic Pleasure–Arousal–Dominance experiments (Valdez and Mehrabian 1994); that is the reason this article measures palettes by lightness L^* and chroma C^* before it says anything about hue. On the learning side, color coding that links a symbol to its referent helps because it

Table 1. Evidence ledger for the color claims used in this article. "Use" says what the article does with the claim; none of the studies listed involved the palettes or figures of this article.

Claim	Source	Status and use
Color–emotion associations are widespread and similar across nations	Jonauskaite et al. (2020); Jonauskaite and Mohr (2025)	Large multinational samples. Prior for palette readings. An association, not an effect on a learner.
Brightness and saturation carry most of the affect (standardized: $P = .69B + .22S$, $A = -.31B + .60S$, $D = -.76B + .32S$)	Valdez and Mehrabian (1994)	One study line. Used only to rank the five palettes (Section 7.3).
Color preference follows the affect of the objects a color evokes	Palmer and Schloss (2010)	Explains why amber can read as warm; not tested here.
Seeing red before a test impairs performance	Elliot et al. (2007); Steele (2014); Gnambs (2020)	Contested. A meta-analysis of 67 effects (38 samples) finds larger effects in early than in later studies. Not used as a design rule.
Round shapes and warm colors raise positive affect; both raise comprehension	Plass et al. (2014)	Closest precedent for separating shape from color. Tested on multimedia lessons; we found no mathematics-specific test.
Color coding directs attention to matching elements and raises retention and transfer	Ozcelik et al. (2009); Mayer (2009)	Used: one color per object, the same in equation and figure.
A color-isolated item is remembered better	Kuhbandner and Pekrun (2013)	Caution: color used for emphasis competes with color used for mood.
Red pens lower the grades graders give	Rutchick, Slepian, and Ferris (2010)	One study, on graders. No design rule taken.
Math anxiety loads working memory; achievement emotions follow control and value appraisals	Ashcraft (2002); Dowker, Sarkar, and Looi (2016); Pekrun (2006)	Reason to treat affect as part of the task, not decoration.
About 8% of men and 0.5% of women have red–green color deficiency	Birch (2012)	Drives the accessibility test of Section 7.2.

shortens the search for corresponding information (Ozcelik et al. 2009), and it is one instance of the signaling principle (Mayer 2009). Shape and color can be separated experimentally: Plass et al. (2014) report that round, face-like shapes and warm colors each induced positive affect and improved comprehension of a multimedia lesson, with transfer improved by round shapes under neutral colors. That is close to what the present construction lets a teacher do with mathematical figures, in a domain (mathematics) where we found no test of it.

What is contested. The best-known "color psychology of achievement" claim, that brief exposure to red impairs test performance (Elliot et al. 2007), is the one a teacher is most likely to meet in professional-development material. A direct replication of a related red/blue priming result failed with more than triple the original sample (Steele 2014), and a meta-analysis of 67 effect sizes from 38 samples found no significant difference for anagram and knowledge tests, lower scores under red for reasoning tests, and effects that shrank from early to later studies (Gnambs 2020). We therefore make no claim of the form "use blue, avoid red." The Pleasure–Arousal–Dominance ranking of Section 7.3 is a prediction about *affect*, not about performance.

Why affect belongs in a mathematics lesson. Math anxiety compromises working memory and pushes learners to avoid mathematics (Ashcraft 2002), and its literature now spans sixty years (Dowker, Sarkar, and Looi 2016). In the control–value account, achievement emotions arise from a learner's appraisals of control and value (Pekrun 2006). If a figure's palette changes those appraisals at all, it is part of the task. Whether it does, for these palettes, is exactly what Section 8 asks.

Who cannot see the palette. About 8% of men and 0.5% of women have red–green color deficiency (Birch 2012). In a class of thirty with an even split that is $0.5 \cdot 30 \cdot 0.08 + 0.5 \cdot 30 \cdot 0.005 \approx 1.3$ students. A palette whose meaning survives only in normal trichromatic vision fails about one student per class; Section 7.2 measures this for all five.

3. The generative field

Let $\rho = \sqrt{x^2 + y^2}$ and $\theta = \operatorname{atan2}(y, x)$. The contour is a symmetry-broken polar curve,

$$r(\theta) = 1 + a\sin(k\theta) + b\sin(m\theta + \phi), \tag{1}$$

where integers k, m set the number of harmonic lobes and the phase ϕ controls how far the curve departs from exact bilateral symmetry. The two terms carry a color each (blue: the k-lobed term; vermillion: the m-lobed term) and keep that color in Figure 1, so that a reader can match term to picture by color alone, as in Ozcelik et al. (2009); the colors are the Okabe–Ito blue and vermillion (Okabe and Ito 2008), chosen so that the pairing survives red–green color deficiency, and the terms are also named in the text so that nothing depends on color alone.

Proposition 3.1 (Symmetry condition)**.** *Let $k \neq m$ be positive integers and $a, b > 0$. Then $r(\pi - \theta) = r(\theta)$ for all θ if and only if k is odd and*

$$\phi \in \begin{cases} \{0, \pi\}, & m \text{ odd}, \\ \{\frac{\pi}{2}, \frac{3\pi}{2}\}, & m \text{ even}. \end{cases} \tag{2}$$

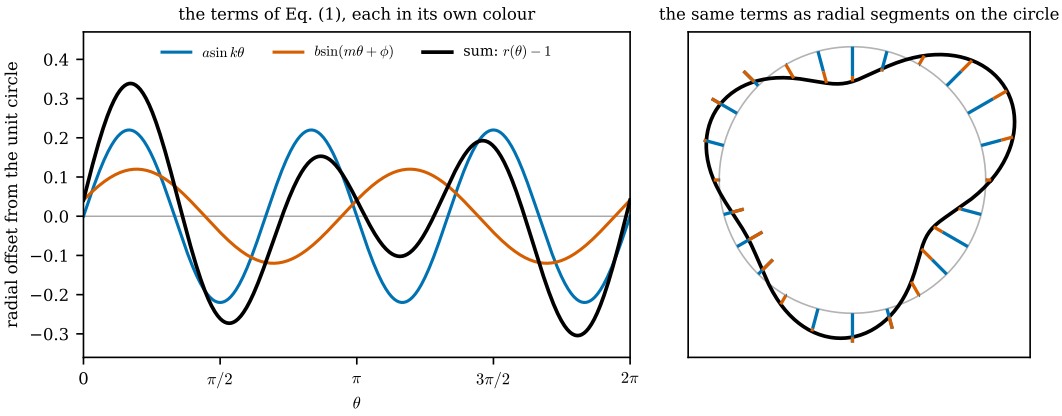

Figure 1. Left: the two terms of Eq. (1) and their sum, as radial offsets from the unit circle, for the Totoro parameters $(k, m, a, b, \phi) = (3, 2, 0.22, 0.12, 0.35)$. Right: the same terms drawn as radial segments on the circle (blue segment, then vermillion segment stacked on it), with the resulting contour in black. Colors match the terms of Eq. (1).

Proof. Expanding $\sin(k(\pi - \theta)) = (-1)^{k+1}\sin(k\theta)$ and $\sin(m(\pi - \theta) + \phi) = (-1)^{m+1}\sin(m\theta - \phi)$, the condition $r(\pi - \theta) = r(\theta)$ for all θ becomes $a(-1)^{k+1}\sin(k\theta) + b(-1)^{m+1}\sin(m\theta - \phi) = a\sin(k\theta) + b\sin(m\theta + \phi)$. Since $k \neq m$, the functions $\sin(k\theta), \sin(m\theta), \cos(m\theta)$ are linearly independent, so coefficients must match separately. The $\sin(k\theta)$ term gives $(-1)^{k+1} = 1$, i.e. k odd. Expanding the m-terms in $\sin(m\theta)$ and $\cos(m\theta)$ and matching coefficients gives $(-1)^{m+1}\cos\phi = \cos\phi$ and $(-1)^{m+1}\sin\phi = -\sin\phi$; for m odd this forces $\sin\phi = 0$, for m even it forces $\cos\phi = 0$. □

Three of the five registers in Section 6 (Totoro, Shishigami, Catbus; Table 3: $k \in \{3, 5\}$) use an odd k together with a ϕ well away from the values Proposition 3.1 identifies as symmetric, so each is a genuine, verified symmetry break, not an accident of rounding. The remaining two (Kodama, No-Face) use an even k instead, for which Proposition 3.1 guarantees *no* phase ϕ restores exact symmetry at all — a stronger, structural asymmetry that Section 6 reads, in each case, against the register's own source material (a rattling, never-still tree spirit; a spirit whose identity the film itself declines to resolve on screen). This is the article's first aesthetic point, stated precisely: perfect bilateral symmetry reads as a stamped logotype, and the viewer's sense that an object is *alive* rather than *printed* is carried by a single continuous parameter, not by an unstructured perturbation.

We checked the "if and only if" by brute force rather than by trusting the proof alone: for every $k, m \in \{1, \ldots, 9\}$ with $k \neq m$ and 41 phases (the four special values $0, \pi/2, \pi, 3\pi/2$ and 37 others), $\max_\theta |r(\pi - \theta) - r(\theta)|$ was computed on a fine grid and compared with the proposition's prediction. All 2,952 of 2,952 cases agree; where the proposition predicts symmetry the largest residual is 1.8×10^{-15} (rounding error), and where it predicts none the smallest residual is 0.004. Figure 2 shows twelve of the cases.

In the classroom. Proposition 3.1 is a parity argument in disguise: $\sin(k(\pi - \theta)) = (-1)^{k+1}\sin(k\theta)$. A student who has met even and odd functions can predict every panel of Figure 2 before seeing it, then use the picture to find the case they got wrong. Polar curves are also a known stumbling block: students' construction of the polar coordinate system is more complicated than the Cartesian one they arrive with

Proposition 1, tested: solid = $r(\theta)$, dashed = its mirror image. 2,952 of 2,952 swept (k, m, ϕ) cases agree with the theorem

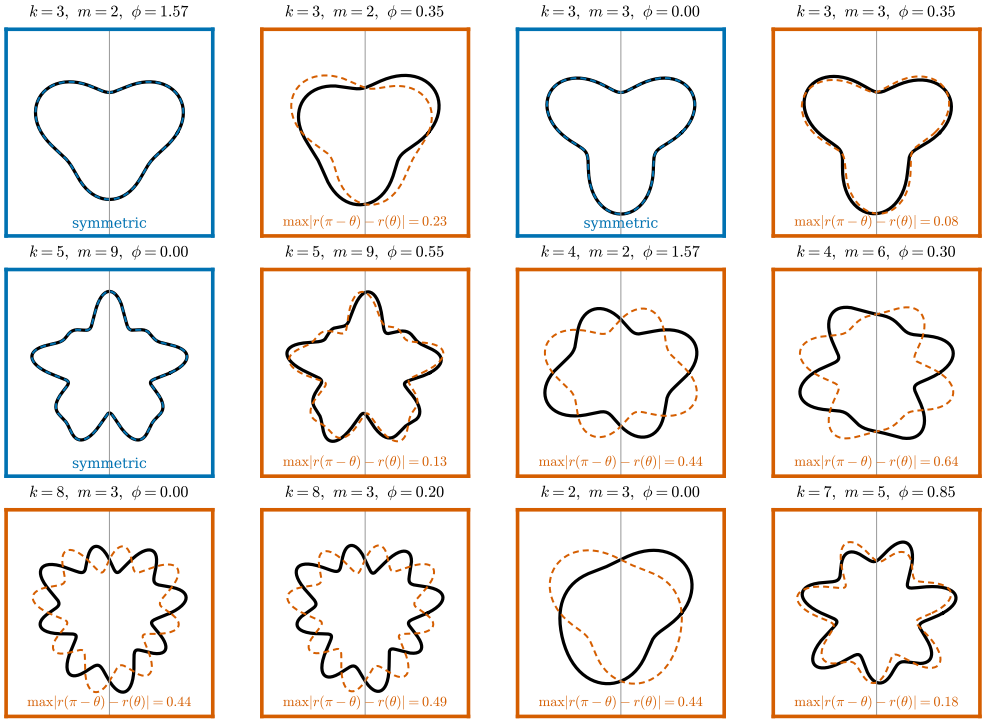

Figure 2. Proposition 3.1 in twelve cases ($a = 0.22$, $b = 0.12$). Solid: $r(\theta)$. Dashed: its mirror image $r(\pi - \theta)$. Blue frames: the curves coincide (symmetric); vermillion frames: they do not, with the maximum gap printed. Every panel with even k ($k = 4, 8, 2$) is asymmetric, including $(k, m, \phi) = (4, 2, \pi/2)$, whose phase would make it symmetric for odd k.

(Moore, Paoletti, and Musgrave 2014), and a decomposition such as Figure 1 shows the radial meaning of $r(\theta)$ before any curve is drawn.

From the contour we build a boundary field and an interior field,

$$B(x, y) = \exp\big(-\lambda\,|\rho - r(\theta)|\big), \tag{3}$$

$$I(x, y) = \frac{1}{1 + \exp\big(\beta(\rho - r(\theta))\big)}, \tag{4}$$

and a metaball field (Blinn 1982) for local mass and fusion,

$$M(x, y) = \sum_{i=1}^{N} \frac{\alpha_i}{(x - x_i)^2 + (y - y_i)^2 + \varepsilon}, \tag{5}$$

where (x_i, y_i) are metaball centers and α_i their signed weights: positive α_i fuse into rounded masses (a belly, a cheek), negative α_i carve hollows (an eye socket, a wound). The full field is

$$\Phi(x, y) = c_B B(x, y) + c_I I(x, y) + c_M \tanh M(x, y) + \eta\, T(x, y), \tag{6}$$

with the texture term T given in Section 5. The rendered image is obtained by applying a color map to Φ and drawing the level sets of interest — nothing more.

3.1. *Excursion sets*

The informal "drawing the level sets of interest" has a precise form. For a threshold u, define the visible body as the excursion set

$$K_u = \{(x, y) \in D : \Phi(x, y) \geq u\}, \tag{7}$$

where $D \subset \mathbb{R}^2$ is the rendering window. Varying u changes what counts as body, ear, tuft, or empty space; the image is an object of level-set geometry, not a fixed bitmap. Section 5 returns to (7) once the texture term T has been defined, to connect it to random level sets.

The simplest place to see the topology of K_u change is the fusion of two metaball wells. Take two equal wells of weight α at $(\pm d/2, 0)$ in (5). On the x-axis, $M(x, 0) = \alpha/((x - d/2)^2 + \varepsilon) + \alpha/((x + d/2)^2 + \varepsilon)$, and the midpoint is a critical point of M by symmetry. Differentiating twice, it is a minimum along the axis exactly when $d^2/4 > \varepsilon/3$, that is $d > 2\sqrt{\varepsilon/3}$ ($= 0.258$ for $\varepsilon = 0.05$); along the y-axis it is a maximum, $M(0, y) = 2\alpha/(d^2/4 + y^2 + \varepsilon)$. So for $d > 2\sqrt{\varepsilon/3}$ the midpoint is a saddle, with value $u_s = 2\alpha/(d^2/4 + \varepsilon)$, and the two blobs of K_u separate exactly when u rises through u_s. For $\alpha = 1$, $\varepsilon = 0.05$ and $d = 1$ this predicts $u_s = 6.667$; scanning u on a 1201×1201 grid and counting connected components, the numerical merge is at 6.688 (0.32% from the prediction, within one scan step of 0.37%; Figure 3).

In the classroom. This is the calculus of two variables stripped to one picture: a critical point, its Hessian, and the fact that a saddle is where level sets change connectivity. A student can move d, watch the merge threshold move, and check the formula with a pencil. The condition $d > 2\sqrt{\varepsilon/3}$ is a nice trap: below it the midpoint is not a saddle and the two wells fuse into one blob at every threshold.

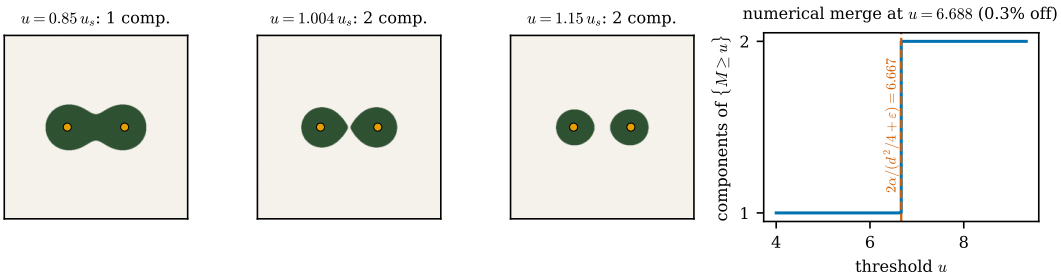

Figure 3. Two equal metaball wells ($\alpha = 1$, $\varepsilon = 0.05$, $d = 1$; orange dots) and the excursion set $\{M \geq u\}$ at $u = 0.85u_s$, $1.004u_s$ and $1.15u_s$ (dark), where $u_s = 2\alpha/(d^2/4 + \varepsilon)$ is the value of M at the midpoint saddle. Right: number of connected components versus u; the dashed line is the analytic saddle value.

4. The geometry of the field

The two ingredients that give the creatures their bodies, the contour (1) and the metaball sum (5), have exact geometry. The lobes of the contour are concavities whose number is fixed by the harmonic orders, and the fusion of two metaball wells is a cusp in the sense of Thom (Thom 1972; Arnold 1992).

4.1. *Convexity, inflections and vertices of the contour*

For a polar curve $\rho = r(\theta)$ with $r > 0$ the signed curvature is

$$\kappa(\theta) = \frac{N(\theta)}{\left(r^2 + r'^2\right)^{3/2}}, \qquad N = r^2 + 2r'^2 - r\,r'', \tag{8}$$

so the curve is convex exactly when $N \geq 0$ everywhere, its inflections are the sign changes of N, and its vertices are the stationary points of κ (do Carmo 1976).

Proposition 4.1 (Convexity of the contour)**.** *Let $r(\theta) = 1 + a\sin(k\theta) + b\sin(m\theta + \phi)$ with $a, b \geq 0$, $a + b < 1$ and $k, m \geq 1$.*

(1) If $a(k^2 + 1) + b(m^2 + 1) < 1$, the contour is a strictly convex closed curve.
(2) If $b = 0$, the contour is convex if and only if $a \leq 1/(k^2 + 1)$. At $a = 1/(k^2 + 1)$ its curvature vanishes exactly at the k points where $\sin k\theta = -1$, and for $a > 1/(k^2 + 1)$ it has exactly $2k$ inflection points.
(3) If $b = 0$, every point with $\cos k\theta = 0$ is a stationary point of the curvature; the contour has at least $2k$ vertices.

Proof. (1) Since $r'^2 \geq 0$, $N \geq r(r - r'')$. Here $r \geq 1 - a - b > 0$ and

$$r - r'' = 1 + a(k^2 + 1)\sin k\theta + b(m^2 + 1)\sin(m\theta + \phi) \geq 1 - a(k^2 + 1) - b(m^2 + 1) > 0,$$

so $N > 0$ and $\kappa > 0$ everywhere.

(2) Put $s = \sin k\theta$. Then $r = 1 + as$, $r'^2 = a^2k^2(1 - s^2)$ and $r'' = -ak^2 s$, so

$$N = 1 + a(k^2 + 2)s + a^2\left(2k^2 + (1 - k^2)s^2\right) =: F(s), \qquad s \in [-1, 1].$$

The coefficient of s^2 is $a^2(1-k^2) \le 0$, so F is concave and its minimum on $[-1,1]$ is attained at an end point. Now $F(1) = (1+a)\big(1+a(k^2+1)\big) > 0$ and

$$F(-1) = (1-a)\big(1-a(k^2+1)\big).$$

Hence $N \ge 0$ everywhere if and only if $a \le 1/(k^2+1)$; at equality N vanishes exactly where $s=-1$. If $a > 1/(k^2+1)$, then $F(-1) < 0 < F(1)$ and the concave function F has exactly one zero $s_0 \in (-1,1)$, at which it changes sign. The equation $\sin k\theta = s_0$ has $2k$ solutions in $[0,2\pi)$, at each of which s crosses s_0, so N changes sign exactly $2k$ times.

(3) With $b=0$, r, r'^2 and r'' are functions of $s = \sin k\theta$, hence so is $\kappa = G(s)$, and $\kappa'(\theta) = G'(s)\,k\cos k\theta$ vanishes wherever $\cos k\theta = 0$; there are $2k$ such points in $[0,2\pi)$. □

For $k=1$ the curve $1 + a\sin\theta$ is a limaçon and part (2) recovers the classical fact that it is convex exactly when $a \le \frac{1}{2}$. For $k \ge 2$, part (3) is a strengthening, for this family, of the four-vertex theorem, which guarantees four vertices on every simple closed convex curve (Mukhopadhyaya 1909; Kneser 1912). The threshold $1/(k^2+1)$ is small: 0.1 for the trefoil $k=3$ and 0.015 for $k=8$. All five registers of Table 3 lie far above it, so every lobe a viewer reads as a cheek, an ear or a limb is a genuine concavity of the silhouette. Evaluating N on the five contours, the inflections number 6 for Totoro and Catbus ($k=3$), 16 for Kodama ($k=8$), and 12 and 18 for No-Face and Shishigami, where the higher-order term $m \in \{6,9\}$ dominates the curvature although its amplitude is smaller: the term of order m enters $r - r''$ with weight $b(m^2+1)$, and $b(m^2+1) > a(k^2+1)$ in both. The anisotropic stretch of Catbus and No-Face (Remark 1) is linear, so it changes neither convexity nor the number of inflections.

4.2. *The fusion of two wells is a cusp*

Section 3.1 computed the threshold at which two equal wells separate. Unequal weights turn that threshold into a curve, and the curve is a cusp. Place two positive wells at $(\pm\delta, 0)$,

$$M(x,y) = \frac{\alpha_1}{(x-\delta)^2 + y^2 + \varepsilon} + \frac{\alpha_2}{(x+\delta)^2 + y^2 + \varepsilon}, \qquad \alpha_1, \alpha_2, \delta, \varepsilon > 0, \tag{9}$$

and write

$$\varphi_\delta(x) = \log\frac{\delta + x}{\delta - x} + 2\log\frac{(x-\delta)^2 + \varepsilon}{(x+\delta)^2 + \varepsilon}, \qquad |x| < \delta. \tag{10}$$

Proposition 4.2 (Fusion of two wells)**.** *(1) Every critical point of M lies on the open segment between the two centres, at a point $(x,0)$ with $\varphi_\delta(x) = \log(\alpha_1/\alpha_2)$.*

(2) If $\delta^2 \le \varepsilon/3$, then M has exactly one critical point, its global maximum, for all weights.

(3) If $\delta^2 > \varepsilon/3$, put

$$x_* = \Big(\delta^2 + \varepsilon - 2\sqrt{\varepsilon(\delta^2+\varepsilon)/3}\Big)^{1/2} \in (0,\delta), \qquad \beta_*(\delta) = -\varphi_\delta(x_*) > 0.$$

If $|\log(\alpha_1/\alpha_2)| < \beta_*(\delta)$*, then* M *has exactly three critical points: two local maxima and a saddle between them. If* $|\log(\alpha_1/\alpha_2)| > \beta_*(\delta)$*, it has exactly one, its global maximum.*

(4) As $\delta^2 \downarrow \varepsilon/3$*,*

$$\beta_*(\delta) = \frac{(\delta^2 - \varepsilon/3)^{3/2}}{\sqrt{2}\,\delta^3}\big(1 + O(\delta^2 - \varepsilon/3)\big).$$

Proof. Write $D_{1,2} = (x \mp \delta)^2 + y^2 + \varepsilon$. Then $\partial_y M = -2y\big(\alpha_1 D_1^{-2} + \alpha_2 D_2^{-2}\big)$ vanishes only on $y = 0$. On the axis, $\partial_x M = 0$ reads

$$\alpha_1 \frac{\delta - x}{\big((x-\delta)^2 + \varepsilon\big)^2} = \alpha_2 \frac{x + \delta}{\big((x+\delta)^2 + \varepsilon\big)^2}.$$

For $x \geq \delta$ the left side is ≤ 0 and the right side positive, and symmetrically for $x \leq -\delta$; so $|x| < \delta$, and taking logarithms gives $\varphi_\delta(x) = \log(\alpha_1/\alpha_2)$. This is (1).

The function φ_δ is odd and tends to $\pm\infty$ as $x \to \pm\delta$. A direct computation gives

$$\varphi_\delta'(x) = -\frac{2\delta\, Q(x^2)}{(\delta^2 - x^2)\big((x-\delta)^2 + \varepsilon\big)\big((x+\delta)^2 + \varepsilon\big)}, \qquad Q(s) = 3s^2 - 6(\delta^2+\varepsilon)s + (3\delta^2 - \varepsilon)(\delta^2+\varepsilon).$$

The roots of Q are $s_\pm = \delta^2 + \varepsilon \pm 2\sqrt{\varepsilon(\delta^2+\varepsilon)/3}$. Always $s_+ > \delta^2$ and $s_- < \delta^2$, since $\varepsilon^2 < \frac{4}{3}\varepsilon(\delta^2+\varepsilon)$; and $s_- > 0$ exactly when $(\delta^2+\varepsilon)^2 > \frac{4}{3}\varepsilon(\delta^2+\varepsilon)$, that is when $\delta^2 > \varepsilon/3$.

If $\delta^2 \leq \varepsilon/3$, then $s_- \leq 0$ and $Q < 0$ on $(0, \delta^2)$, so φ_δ is strictly increasing on $(-\delta, \delta)$ and takes every value exactly once: M has one critical point. As $M > 0$ and $M \to 0$ at infinity, it is the global maximum. This is (2).

If $\delta^2 > \varepsilon/3$, then $x_* = \sqrt{s_-} \in (0, \delta)$, φ_δ decreases on $[-x_*, x_*]$ from β_* to $-\beta_*$ and increases on $(-\delta, -x_*]$ and $[x_*, \delta)$, from $-\infty$ to β_* and from $-\beta_*$ to $+\infty$. Hence $\varphi_\delta(x) = c$ has three solutions for $|c| < \beta_*$ and one for $|c| > \beta_*$. Along $y = 0$ the restriction of M is positive with limit 0 at $\pm\infty$, so its three critical points are a maximum, a minimum and a maximum, in this order; and $\partial_y^2 M = -2\big(\alpha_1 D_1^{-2} + \alpha_2 D_2^{-2}\big) < 0$ on the axis, so the middle one is a saddle of M. This is (3).

For (4) put $t = \delta^2 - \varepsilon/3$. Since φ_δ is odd, $\varphi_\delta(x) = \varphi_\delta'(0)\,x + \frac{1}{6}\varphi_\delta'''(0)\,x^3 + O(x^5)$, and from the formula for φ_δ',

$$\varphi_\delta'(0) = \frac{2(\varepsilon - 3\delta^2)}{\delta(\delta^2+\varepsilon)} = -\frac{3t}{2\delta^3}\big(1 + O(t)\big), \qquad \varphi_\delta'''(0) = \frac{6}{\delta^3}\big(1 + O(t)\big).$$

The minimum of φ_δ on $(0, \delta)$ is therefore attained at $x_*^2 = -2\varphi_\delta'(0)/\varphi_\delta'''(0)\,(1 + O(t)) = \frac{t}{2}(1 + O(t))$, and $\beta_* = -\frac{2}{3}\varphi_\delta'(0)\,x_*\,(1 + O(t)) = t^{3/2}/(\sqrt{2}\,\delta^3)\,(1 + O(t))$. □

In the plane of the two control parameters, the separation δ and the log-ratio $\log(\alpha_1/\alpha_2)$ of the weights, the set where the field has two bodies is bounded by the curve $|\log(\alpha_1/\alpha_2)| = \beta_*(\delta)$, which leaves the point $(\sqrt{\varepsilon/3}, 0)$ as a semicubical cusp (Figure 4). This is Thom's cusp catastrophe (Thom 1972; Arnold 1992): at $\delta^2 = \varepsilon/3$ and $\alpha_1 = \alpha_2$ the midpoint is a degenerate maximum of M along the axis, of the form $-x^4$, and the two parameters unfold it as $x^4 + ux^2 + vx$, with u proportional to $\varepsilon/3 - \delta^2$ and v to $\log(\alpha_1/\alpha_2)$. For the separation $2\delta = 1$ and $\varepsilon = 0.05$ of Section 3.1, Proposition 4.2 gives $\beta_* = 3.128$: the two wells remain two bodies at every threshold

between the saddle value and the smaller maximum as long as the heavier weight is less than $e^{3.128} \approx 22.8$ times the lighter one, and beyond that the lighter well is absorbed into the flank of the heavier one at every threshold. A path in the control plane that crosses the cusp away from its vertex makes a body appear or vanish abruptly; a path through the vertex makes one body divide into two. In the language of morphogenesis that Thom proposed for such pictures, the egg hatches along the cusp.

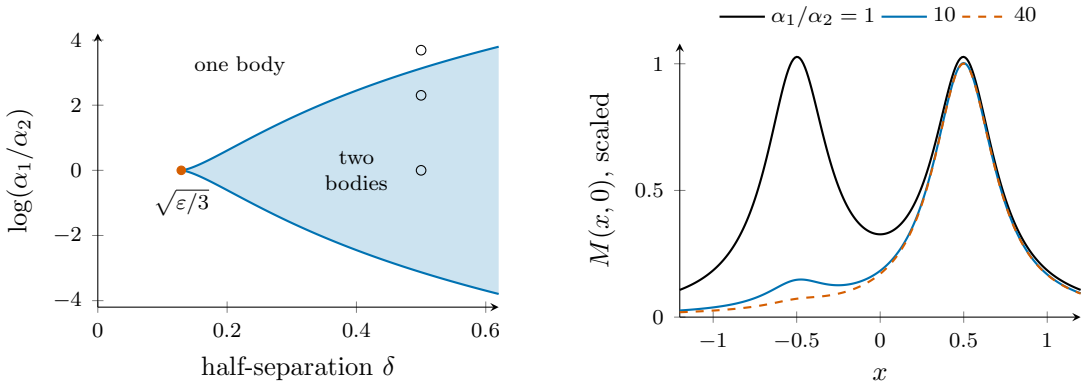

Figure 4. The cusp of Proposition 4.2 for $\varepsilon = 0.05$. Left: the shaded region, bounded by $\pm\beta_*(\delta)$, is where the two wells give two bodies; its vertex is $(\sqrt{\varepsilon/3}, 0)$. Circles: the three weight ratios of the right panel at $2\delta = 1$. Right: the field on the axis through the centres, scaled to a common maximum. At the ratios 1 and 10 there are two maxima and a saddle; at $40 > e^{\beta_*} = 22.8$ the lighter well is a shoulder on the heavier one.

Topology of the excursion sets. Proposition 4.2 is the two-well case of a general principle: the excursion sets $K_u = \{\Phi \geq u\}$ of (7) change their topology only when u crosses a critical value. As u decreases, a maximum creates a component, a saddle either joins two components or encloses a hole, and a minimum fills a hole; for a field without degenerate critical points whose maxima, saddles and minima lie inside the window, the Euler characteristic of K_u is the number of maxima minus the number of saddles plus the number of minima at levels at least u (Milnor 1963). The counts b_0 and χ of Figure 9 are this principle observed on the full field.

5. Texture as a wavelet field

A single sinusoid is the wrong tool for fur or moss: neither is periodic at one frequency, and a one-centimeter patch and a one-meter patch of moss look statistically similar without either looking like a pure sine wave. This is precisely the situation wavelets were built for: unlike a Fourier mode, a wavelet is a small bump localized in both space and scale, and a sum of dilated, translated copies represents roughness at every scale simultaneously. Our construction is a simplified relative of wavelet noise (Cook and DeRose 2005), built on the periodized Haar mother wavelet (Haar 1910), $\psi : \mathbb{R} \to \{-1, +1\}$,

$$\psi(t) = \begin{cases} +1, & \{t\} \in [0, \tfrac{1}{2}), \\ -1, & \{t\} \in [\tfrac{1}{2}, 1), \end{cases} \tag{11}$$

where $\{t\}$ is the fractional part of t. Fixing a base frequency $f_0 > 0$, a range of octaves $j = 0, \ldots, J$, and a roughness (Hurst-type) exponent $H \in (0, 1)$, the wavelet texture

field is

$$T(x,y) = \frac{1}{Z}\sum_{j=0}^{J} 2^{-jH}\, c_j(x,y)\, \psi\big(f_0 2^j x\big)\, \psi\big(f_0 2^j y\big), \tag{12}$$

where $c_j(x,y) \in [-1,1]$ is constant on each dyadic cell and Z rescales T to a fixed range. Reading (12) as an inverse discrete wavelet transform: T is reconstructed from coefficients c_j living at dyadic scales $2^{-j}/f_0$, each contributing amplitude 2^{-jH}. Small j (coarse scale) reads as moss patches; large j (fine scale) reads as individual fur strands. H is the same roughness exponent used in wavelet syntheses of fractional Brownian motion (Flandrin 1992): H near 1 favors smooth, coherent patches, H near 0 favors sharp, high-frequency detail — soft moss against hirsute fur, tuned with one number. This is the mechanism behind one half of Section 6's No-Face register: its texture uses a markedly lower H and more octaves J than the Totoro register, producing a visibly finer, rougher weave that we return to, quantitatively, in Section 7.

5.1. *Multiresolution structure*

Equation (12) is not an ad hoc sum of sinusoids at dyadic frequencies; it is a truncated instance of a multiresolution analysis (MRA) in the sense of Mallat's synthesis of the subject (Mallat 2008): a nested sequence of approximation spaces $V_0 \subset V_1 \subset \cdots$, generated by dyadic dilates and integer translates of one scaling function, whose successive orthogonal complements $W_j = V_{j+1} \ominus V_j$ are spanned by dyadic dilates and translates of one mother wavelet. The periodized Haar function ψ used here is, in this language, the simplest possible compactly supported orthonormal wavelet: its scaling function is the indicator of $[0,1)$, and V_0 is exactly the space of functions constant on unit cells. Working at finitely many octaves $j = 0, \ldots, J$, rather than the full MRA limit $J \to \infty$, is precisely what keeps T a finite sum evaluable at every grid point of Algorithm 1, at the cost of a texture that is band-limited rather than genuinely fractal; Flandrin (1992)'s fractional-Brownian-motion synthesis is the same truncation carried to its own infinite-octave limit.

Two further points about the wavelet itself, not just the octave sum, are worth making precise. First, $\psi(f_0 2^j x)\,\psi(f_0 2^j y)$ in (12) is a *separable* two-dimensional wavelet: a plain tensor product of the same one-dimensional ψ along each axis, not one of the genuinely two-dimensional (e.g. isotropic or directional) constructions the wavelet literature also studies. Separability is what makes CellHash in Algorithm 1 a per-cell scalar rather than a per-orientation vector: the price is a texture with a faint axis-aligned bias, visible on close inspection as the diagonal checkerboard grain in Figures 6 and 18 (most evident in Kodama's flat, low-η background, where nothing else competes with it for attention). Second, the Haar wavelet has exactly one vanishing moment ($\int \psi = 0$, but $\int t\,\psi(t)\,dt \neq 0$ in general), the minimum possible for an orthonormal wavelet and the reason ψ is a discontinuous square wave rather than a smooth bump. Daubechies's construction of compactly supported orthonormal wavelets with an arbitrary number of vanishing moments (Daubechies 1988) is the standard way to trade this discontinuity for smoothness: more vanishing moments annihilate more polynomial content locally, producing a smoother ψ at the cost of wider support. We did not make that trade. Haar's own discontinuity is the point: fur, moss, and scale boundaries are themselves discontinuous at the scale this article renders them, and a smoother mother wavelet would soften exactly the roughness Section 6 relies on H to control.

Figure 6 isolates this effect, rendering T alone, with f_0 and J fixed, at the five H values actually used by the registers of Section 6. The visual difference between panels is real but subtle, because every octave $j = 0, \dots, J$ contributes at every H, only reweighted by 2^{-jH}; Figure 7 makes that reweighting explicit, plotting 2^{-jH} against j for each H: at $H = 0.88$ (Kodama) the weight has collapsed to under 3% of its $j = 0$ value by $j = 6$, so the field is effectively coarse; at $H = 0.45$ (No-Face) it retains over 15% of that value at the same octave, so fine scales remain visibly load-bearing. This is the precise sense in which H is a one-number dial between moss and matter that has not finished resolving.

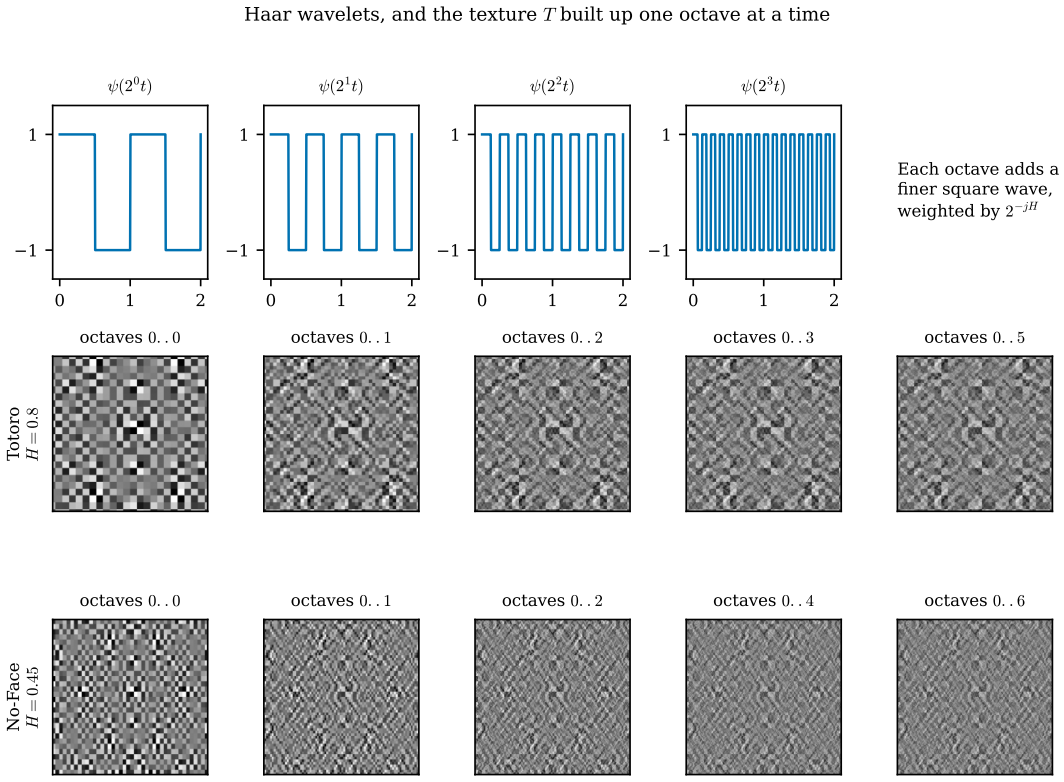

Figure 5. Top: the Haar mother wavelet dilated by 2^j, $j = 0, \dots, 3$. Below: the texture T of (12) truncated after octave J', for the Totoro ($H = 0.8$) and No-Face ($H = 0.45$) parameters (f_0, J, H) of Table 3. At $H = 0.45$ the later octaves keep changing the picture; at $H = 0.8$ they barely do.

In the classroom. The sum (12) is a geometric series in disguise: octave j is weighted by r^j with $r = 2^{-H}$, and the fraction of the total weight left after octave J' is a one-line computation a student can do before running anything. Figure 5 then lets them see what that number means, and the choice between $H = 0.8$ and $H = 0.45$ turns "convergence rate of a series" into "moss or matter."

That reweighting has a further consequence worth checking directly rather than taking on faith: if the octave terms $U_j = 2^{-jH} c_j(x, y)\, \psi(f_0 2^j x)\, \psi(f_0 2^j y)$ of (12) were mutually uncorrelated over the sampled grid, the variance of $T = Z^{-1} \sum_j U_j$ would split additively, $\mathrm{Var}(T) = Z^{-2} \sum_j \mathrm{Var}(U_j)$, with no cross terms — a Parseval-type energy accounting familiar from the full translated Haar system $\psi_{j,k}(t) = 2^{j/2} \psi(2^j t - k)$, whose orthogonality is classical. Our sum is not literally an instance of that theorem: it fixes phase and sums only over scale j, not over every integer shift k, and c_j is a

deterministic pseudorandom hash rather than a probabilistically independent coefficient, on the finite, non-periodic domain $[-2.2, 2.2]^2$ Algorithm 1 actually samples — so additivity here is a plausible approximation, not a corollary. We checked it numerically on the real per-register (f_0, J, H) of Table 3: the additive prediction matches the true sampled variance to within 0.005–0.06% across all five registers (Table 2), which lets us read off each octave's real share of the texture's variance rather than only the qualitative 2^{-jH} curve of Figure 7. Kodama ($H = 0.88$) puts 61.7% of its texture variance in the coarsest octave alone, with under 1% left at $j = 4$; No-Face ($H = 0.45$) puts 39.8% at $j = 0$ and still carries 1.3% at its finest octave $j = 6$ — the same smooth-versus-rough contrast Section 6 argues for qualitatively, now attached to numbers computed from the actual field rather than asserted from the shape of 2^{-jH} alone.

Table 2. Empirical Parseval check and per-octave variance share of the wavelet texture T, computed from the real (f_0, J, H) of each register (seed fixed at 7, matching every other figure in this article). "rel. err." is $|\operatorname{Var}(T) - Z^{-2}\sum_j \operatorname{Var}(U_j)|/\operatorname{Var}(T)$; the remaining columns are each octave j's share of $\sum_j \operatorname{Var}(U_j)$, in %.

Register	rel. err.	j=0	j=1	j=2	j=3	j=4	j=5	j=6
Kodama	0.005%	61.7	27.0	8.1	2.5	0.7	—	—
Totoro	0.061%	59.5	27.0	9.0	3.1	1.0	0.3	—
Catbus	0.052%	58.8	27.1	9.4	3.3	1.1	0.4	—
Shishigami	0.013%	48.9	29.0	12.5	5.6	2.4	1.1	0.5
No-Face	0.020%	39.8	28.5	15.3	8.4	4.4	2.4	1.3

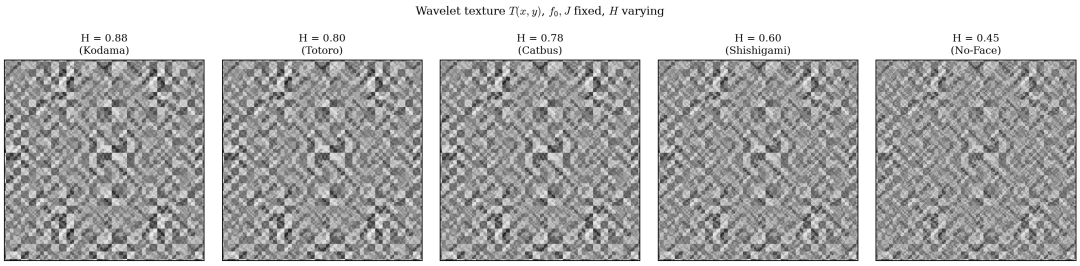

Figure 6. The wavelet texture field $T(x, y)$ in isolation, for the five roughness exponents H used in Section 6, with f_0 and J fixed.

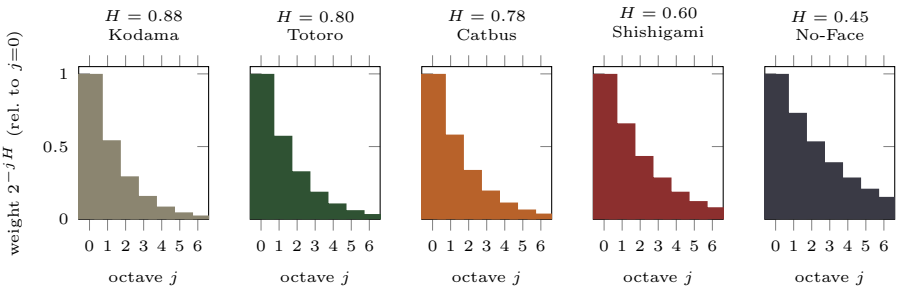

Figure 7. Per-octave weight 2^{-jH}, relative to $j = 0$, for the five roughness exponents used in Section 6 (native `pgfplots`; values computed exactly, not read off a rendered image). Lower H retains more weight at high octaves (fine scale), which is the mechanism, not merely the appearance, of a rougher texture in Figure 6.

5.2. *A stochastic remark*

Replacing the deterministic coefficients $c_j(x, y)$ in (12) with a random field turns T, and hence Φ, into a random field on $\mathbb{R}^2$; the excursion set (7) at threshold u becomes the random compact set $K_u(\omega)$, connecting this construction to the standard machinery of random level sets. A moss-and-lichen forest floor is, in exactly this sense, one realization of a random field. The canonical object in that theory is the Gaussian free field (Sheffield 2007), whose point-to-point correlations decay logarithmically with distance rather than exponentially: if the wavelet texture above is a deliberately simple model of a forest floor, the Gaussian free field is what that same floor would look like if correlation were demanded at every scale at once, with no artificial cutoff between octaves j. We record it because it shows the construction of Section 3 sits inside, rather than beside, an existing mathematical theory, and we take the connection one step further below.

One step further: for a smooth, sufficiently regular random field on a domain $D \subset \mathbb{R}^2$ — in particular for an isotropic Gaussian field, which the Gaussian free field itself is not, but genuinely smoother relatives of it are — the Gaussian kinematic formula gives the expected Euler characteristic of the excursion set (7) as an explicit function of the threshold u and of the field's covariance structure alone (Adler and Taylor 2007):

$$\mathbb{E}\big[\chi(K_u)\big] = \sum_{j=0}^{2} \mathcal{L}_j(D)\, \rho_j(u), \tag{13}$$

where $\mathcal{L}_j$ are the Lipschitz–Killing curvatures of the rendering window D (for a planar D, $\mathcal{L}_2$ is its area, $\mathcal{L}_1$ half its perimeter, $\mathcal{L}_0$ its Euler characteristic) and ρ_j are explicit Gaussian-derived densities depending on u and on the field's spectral moments. Our own deterministic-hash texture is not Gaussian, so (13) does not apply to it literally; what it identifies is exactly what a fully stochastic version of the construction would buy over the present one: not only a picture of one excursion set, as in Figure 8, but a closed-form prediction for how many connected components, holes, and how much total boundary length to expect at threshold u, without simulating a single realization. The qualitative distinction Table 3 draws between "gentle, rounded" and "pale, near-still" is, in this precise sense, a claim about the sign and size of $\rho_1(u)$ and $\rho_2(u)$ at the thresholds each register actually uses — a quantitative question the present article does not settle, but that (13) makes precise enough to settle.

The hash $c_j(x, y)$ actually used in (12) is deterministic given a seed ω, but the seed itself indexes a family of independent draws: different ω give different, uncorrelated realizations T_ω of the same texture field, in exactly the sense the paragraph above describes. Figure 8 makes this concrete, rendering three independent seeds for each of the five registers of Section 6 under identical parameters and color maps. Two things are visible at once: the local, fine-grained detail of the fur, scales, or skin changes from column to column, exactly as a different draw of a random field should; and the coarse shape and register-defining reading of each row does not, because that reading is carried by B, I, M and the palette, not by T. This is also the empirical warrant for treating each register in Section 6 as a reading of the construction rather than a single fortunate rendering: three independent realizations of the field per register agree on the reading, so it is not an artifact of one particular seed.

Figure 8. Three independent texture realizations T_ω (seeds 3, 7, 15) for each of the five registers, at otherwise identical parameters. Fine-scale detail varies across columns; the coarse shape and register-defining reading of each row is stable across the same columns.

5.3. *The gradient vector field and boundary flow*

With T now in hand, $\Phi = c_B B + c_I I + c_M \tanh M + \eta T$ of (6) is complete, and it has an associated vector field, its gradient $\nabla\Phi = (\partial_x\Phi, \partial_y\Phi)$, zero exactly at critical points and otherwise pointing in the direction of steepest increase. Away from critical points, the excursion-set boundary $\partial K_u = \{\Phi = u\}$ is a smooth curve orthogonal to $\nabla\Phi$ at every point: $\nabla\Phi$ is the outward normal of the creature's own silhouette, not merely an auxiliary quantity.

Proposition 5.1 (Threshold flow)**.** *Fix (x_0, y_0) with $\Phi(x_0, y_0) = u_0$ and $\nabla\Phi(x_0, y_0) \neq 0$. There is a unique smooth curve $u \mapsto \mathbf{x}(u)$ defined near u_0, with $\mathbf{x}(u_0) = (x_0, y_0)$, $\Phi(\mathbf{x}(u)) = u$, and $\mathbf{x}'(u)$ parallel to $\nabla\Phi(\mathbf{x}(u))$; it satisfies*

$$\mathbf{x}'(u) = \frac{\nabla\Phi(\mathbf{x}(u))}{|\nabla\Phi(\mathbf{x}(u))|^2}. \tag{14}$$

Proof. Existence and smoothness of $\mathbf{x}(u)$ follow from the implicit function theorem applied to $F(\mathbf{x}, u) = \Phi(\mathbf{x}) - u$, since $\nabla\Phi(x_0, y_0) \neq 0$. Differentiating $\Phi(\mathbf{x}(u)) = u$ in u gives $\nabla\Phi(\mathbf{x}(u)) \cdot \mathbf{x}'(u) = 1$. Requiring $\mathbf{x}'(u)$ parallel to $\nabla\Phi$ — pure normal motion, no tangential drift along the level curve — pins down $\mathbf{x}'(u) = c\,\nabla\Phi(\mathbf{x}(u))$ for a scalar c; substituting into the constraint gives $c = 1/|\nabla\Phi(\mathbf{x}(u))|^2$. □

Equation (14) is the same normal-velocity law underlying level-set methods for propagating fronts generally: raising the threshold u pushes the boundary along $\nabla\Phi$, at a rate inversely proportional to how steeply the field is already changing there. Where Φ is steep — near the sharp ridge of the boundary field B, Eq. (1) — the silhouette barely moves as u varies, which is why every register in Section 6 reads as one stable outline rather than a family of visibly different silhouettes at nearby thresholds. Where Φ is shallow — deep inside a metaball well, or in the low-amplitude texture-dominated background of (12) — the same change in u sweeps the boundary a long way; this is the mechanism, not merely the presentation choice, behind Section 6's multiple-contour-level figures (Shishigami's paired day/night boundary, No-Face's mask-and-hollow pair): different thresholds of one flow, not different fields.

Equation (14) is a first-order statement, so the right test is how its error scales with the step, not whether one error is small. On the smooth part of the field ($\eta = 0$; the Haar texture is piecewise constant, and its gradient is supported on cell edges), for 360 boundary points sampled from the Totoro, Shishigami and Catbus fields at three thresholds each, we moved the level from u to $u + \Delta u$ by Newton root-finding on a cubic-spline interpolant and compared the *normal* displacement with $\Delta u\,|\nabla\Phi|^{-1}$. The median relative error is 6.13% at $\Delta u = 4\%$ of the field's range, 1.56% at 1% and 0.39% at 0.25%: linear in Δu, as a first-order law must be. The 95th percentile is larger (62.23%, 15.07%, 4.53%) and shrinks at the same rate; we did not isolate which boundary points carry the tail, and a first-order law is expected to fail first where the level curve bends sharply (Figure 9).

In the classroom. The proposition is the implicit function theorem with a picture attached, and the check is a lesson in numerical analysis: a first-order method has first-order error, and one should test a scaling law by changing the step, not by looking at one number. The Euler characteristic $\chi(K_u)$ in Figure 9 is the quantity whose expected value the Gaussian kinematic formula (13) predicts for a random field; here it is computed for one realization by counting.

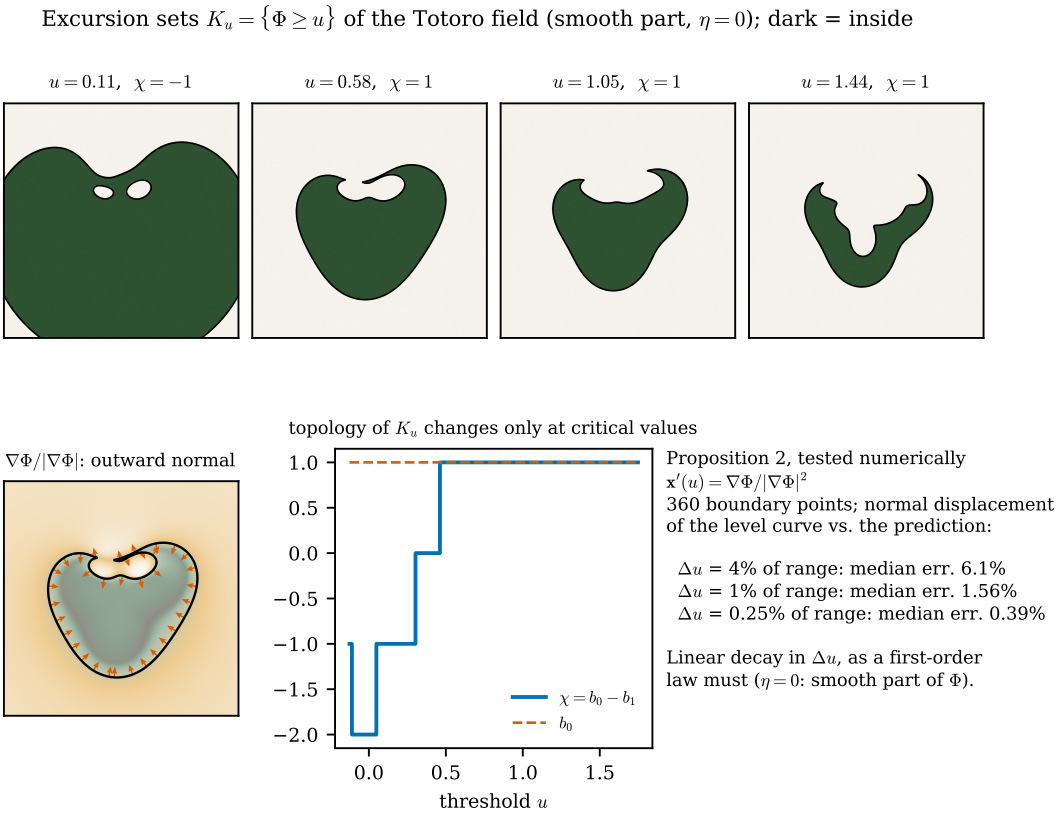

Figure 9. Top: excursion sets K_u of the smooth Totoro field at four thresholds, with Euler characteristic $\chi = b_0 - b_1$ (components minus holes). Bottom left: the outward normal $\nabla\Phi/|\nabla\Phi|$ on one level curve. Bottom middle: b_0 and χ as functions of u; they change only at critical values. Bottom right: the numerical check of Proposition 5.1. Components under 40 pixels are discarded (the tanh saturation makes the top of the field a numerical plateau).

The same $\nabla\Phi$ does double duty in every rendered figure in this article. Treating Φ as a heightfield and $\nabla\Phi$ as a surface normal's tangent components gives a standard cartographic hillshade: a fixed light direction dotted against $(-\nabla\Phi, 1)/\|(-\nabla\Phi, 1)\|$ produces the subtle relief visible in every creature figure from here on (most clearly in Figures 12–17 and 18) — moss-patch and fur-strand texture reading as raised or recessed, not merely light or dark. This is a presentation choice, disclosed as such, layered on top of the same color map; it adds no new mathematics beyond $\nabla\Phi$, already in hand from Proposition 5.1 above.

5.4. *Algorithmic summary*

Sections 3–5's construction is elementary once stated as an algorithm: evaluate four explicit scalar functions on a grid and combine them linearly, after one tanh. Algorithm 1 states it in full; every figure of the creatures is this computation, and nothing else, followed by a color map.

Algorithm 1 The generative field Φ on a grid

Require: grid resolution N, extent E, contour parameters $(a, b, k, m, \phi, \lambda, \beta)$, metaball centers/weights $\{(x_i, y_i, \alpha_i)\}$, wavelet parameters (f_0, J, H), weights (c_B, c_I, c_M, η), seed ω
Ensure: field values Φ_{jk} for grid points $(x_j, y_k) \in [-E, E]^2$

for $j, k \leftarrow 1$ **to** N **do**
  $\rho \leftarrow \sqrt{x_j^2 + y_k^2}$; $\theta \leftarrow \operatorname{atan2}(y_k, x_j)$
  $r \leftarrow 1 + a\sin(k\theta) + b\sin(m\theta + \phi)$ ▷ Eq. (1)
  $B \leftarrow \exp(-\lambda|\rho - r|)$; $I \leftarrow 1/(1 + \exp(\beta(\rho - r)))$
  $M \leftarrow \sum_i \alpha_i/((x_j - x_i)^2 + (y_k - y_i)^2 + \varepsilon)$ ▷ Eq. (5)
  $T \leftarrow 0$
  for $\ell \leftarrow 0$ **to** J **do**
    $s \leftarrow f_0 \cdot 2^\ell$
    $c \leftarrow \textsc{CellHash}(\lfloor sx_j \rfloor, \lfloor sy_k \rfloor, \ell, \omega)$ ▷ deterministic given ω; Sec. 5
    $T \leftarrow T + 2^{-\ell H}\, c \cdot \psi(sx_j)\, \psi(sy_k)$
  end for
  $\Phi_{jk} \leftarrow c_B B + c_I I + c_M \tanh(M) + \eta T$
end for

6. Five registers of one construction

Table 3 gives the five parameter regimes. Only the harmonic orders k, m carry a fixed, deliberate meaning across regimes: low, closely spaced orders read as a gently lobed body; high, widely spaced orders read as sharp, irregular protrusions; and, by Proposition 3.1, an even k (Kodama, No-Face) admits no phase ϕ under which the contour is exactly symmetric, at any amplitude. All other continuous parameters (contour amplitudes, metaball centers and weights, wavelet f_0, J, H, and, for Catbus and No-Face, the anisotropy parameter introduced below) are concrete, disclosed choices consistent with each register's qualitative design intent, not a reconstruction of a specific prior numerical implementation.

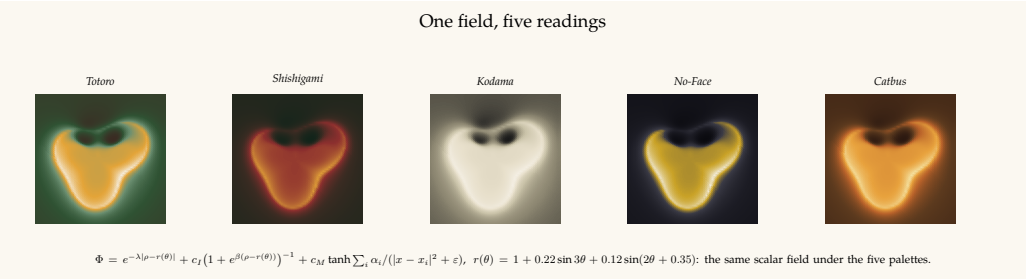

Figure 10. One field, five readings: the Totoro field $c_B B + c_I I + c_M \tanh M$ of Eq. (6), without texture, under the five palettes of Section 7.

Table 3. The five parameter regimes rendered in Figure 18. (k, m): contour harmonic orders. Metaballs: number of wells and sign pattern. H: wavelet roughness exponent (lower = rougher).

Register	(k, m)	Metaballs	H	Reading
Totoro	$(3, 2)$	5, mostly positive	0.80	gentle, rounded
Shishigami	$(5, 9)$	3, two strongly negative	0.60	ambivalent, antlered
Kodama	$(8, 3)$	3, weakly positive	0.88	pale, near-still
No-Face	$(4, 6)$	3, one strongly positive	0.45	void-bodied, single accent
Catbus	$(3, 2)$	7, mixed positive	0.78	warm, wide

Remark 1 (An anisotropic extension)**.** Catbus and No-Face are built from an ambient coordinate $(x/\alpha, y)$ rather than (x, y) directly, for an anisotropy factor α (1.55 for Catbus's wide, bus-like silhouette; 0.72 for No-Face's taller, robed one) substituted into every subsequent step (contour, metaballs, texture). Since scaling only the x-coordinate commutes with the reflection $x \mapsto -x$ underlying Proposition 3.1's $\theta \mapsto \pi - \theta$ symmetry test, the anisotropic registers satisfy exactly the same symmetry condition on (k, m, ϕ) as the isotropic ones; α changes the silhouette's aspect ratio without touching which (k, m, ϕ) triples are symmetric.

Remark 2 (A quasicrystal aside)**.** The harmonic orders in Table 3 are not evenly drawn from the positive integers. Totoro's and Catbus's pair, and No-Face's k, lie in $\{2, 3, 4\}$; Shishigami reaches into $\{5, 9\}$, and Kodama's $k = 8$ reaches further still. That split tracks a genuine mathematical boundary. The crystallographic restriction theorem, classical in the theory of periodic tilings of the plane, permits rotational symmetry only of orders $1, 2, 3, 4, 6$: a periodic pattern can never have exact 5-fold, 8-fold, or 9-fold rotational symmetry, on pain of contradiction with translational periodicity. Such non-crystallographic orders were long thought physically unrealizable for exactly this reason, until Shechtman and coauthors reported a real metallic phase with long-range icosahedral (and hence non-crystallographic) orientational order and no translational symmetry at all — a quasicrystal — (Shechtman et al. 1984), work recognized by the 2011 Nobel Prize in Chemistry; octagonal (8-fold) quasicrystalline phases matching Kodama's own k have since been reported as well. Shishigami's $(5, 9)$ and Kodama's $k = 8$ are drawn from exactly this forbidden set; Totoro's, No-Face's, and Catbus's harmonic orders are not. This is not a mechanism the article claims to model, and no aperiodic tiling appears anywhere in the construction; it is offered as a reading, not a derivation: the same integers excluded from every periodic pattern in the plane are, here, the ones carrying Shishigami's hostility and Kodama's restlessness — two registers whose viewers read them as anything but a stamped logotype, for reasons this remark does not claim are the only ones available, only that the excluded

integers and the unsettled reading happen to coincide.

Before the five registers, Figure 11 shows what the two harmonic orders alone do. It renders the Totoro field, unchanged except for (k, m), over a 4×4 grid of values; the diagonal $k = m$ is excluded because Proposition 3.1 requires $k \neq m$. It is a parameter atlas a student can extend by editing two integers.

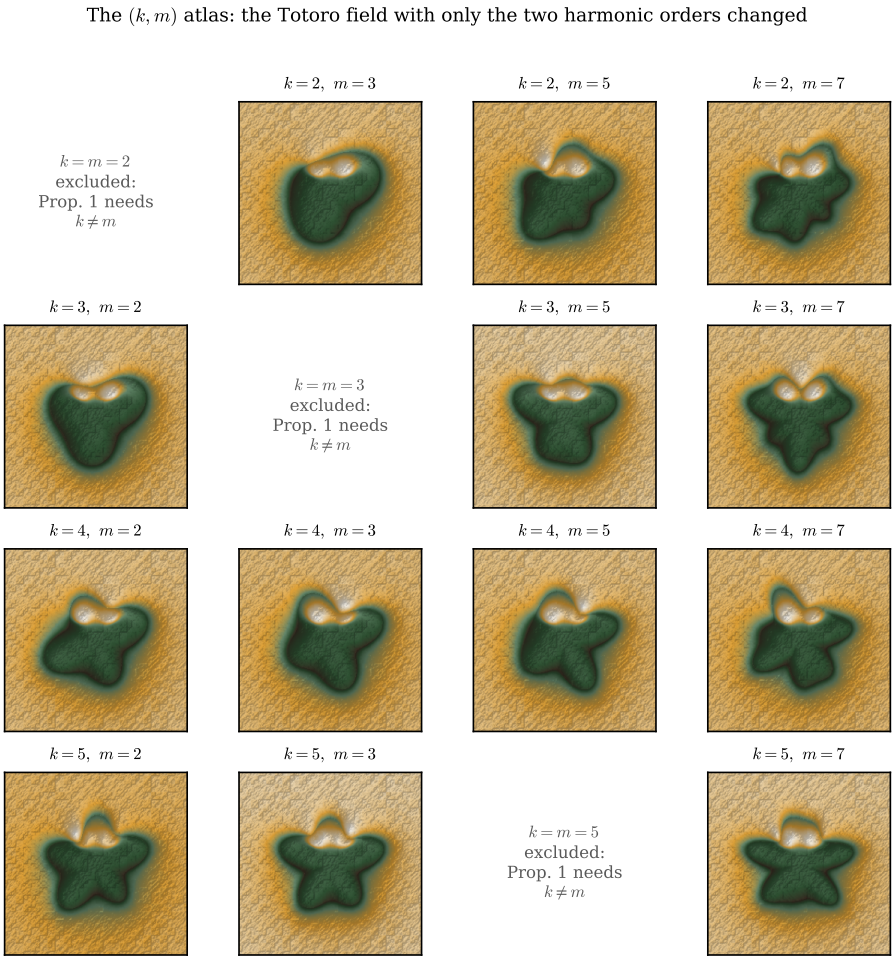

Figure 11. The (k, m) atlas: the Totoro field ($\phi = 0.35$, all other parameters and the palette fixed) for $k \in \{2, 3, 4, 5\}$ and $m \in \{2, 3, 5, 7\}$.

Figure 18 renders all five. The Totoro register — named for the gentle forest spirit of Hayao Miyazaki's 1988 film *My Neighbor Totoro*, an early and central touchstone for how the present construction was designed to read as tender rather than merely round — has five mostly-positive metaballs that fuse into a single rounded torso with two merged, light-toned lobes read as sleepy eyes; the low harmonic orders $(3, 2)$ keep the silhouette close to a soft, breathing curve rather than a rigid outline.

The Shishigami register — Miyazaki's antlered forest god from *Princess Mononoke* (1997), who presides over life and death within the same body and whose day form is placid while his night form, the Nightwalker, is a towering, faceless silhouette — uses higher harmonic orders to produce an antlered, star-like body, and its two strongly negative metaballs hollow out dark, illegible eye sockets: sacred but never quite safe, matching the character's own double nature.

Figure 12. The Totoro register on its own. A warm radial vignette, added only in this enlarged rendering, darkens the corners toward a contained, held warmth rather than an open field; field and palette are unchanged from Table 3 and Figure 18.

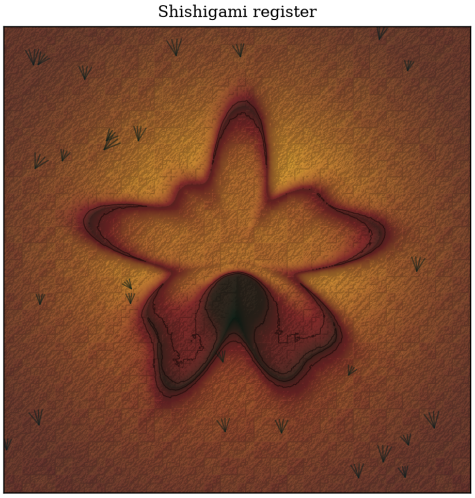

Figure 13. The Shishigami register on its own. Two contour levels, rather than the single level used for Totoro, are drawn together to give a paired day/night reading: an outer boundary close to the placid day form and an inner one hollowing toward the illegible night form, with no vignette added. Field and palette are exactly those of Table 3 and Figure 18.

The Kodama register, named for the tree spirits of *Princess Mononoke* (1997), draws on folklore that long predates the film: kodama are, in Shinto-inflected tree veneration traced to the eighth-century *Kojiki* and *Nihon Shoki*, held to be the animating spirits of trees old enough to have accumulated one, dying if their tree dies, and conventionally illustrated (a design choice the film follows) as small, pale, dual-toned bobble-heads that visibly rattle rather than sit still. Miyazaki's own animism has been read, in recent film scholarship, as more than decoration: it stages nature with its own agency rather than as a passive backdrop, kodama among the clearest cases (Yoneyama 2021). The register built for them keeps the metaballs weak and the wavelet roughness the highest of the five ($H = 0.88$, Table 3), for a nearly flat, matte surface rather than fur or tissue; only the even harmonic order $k = 8$, guaranteeing perpetual asymmetry by Proposition 3.1, keeps the silhouette from settling into a still circle, tracking the character's own restlessness rather than its gentleness.

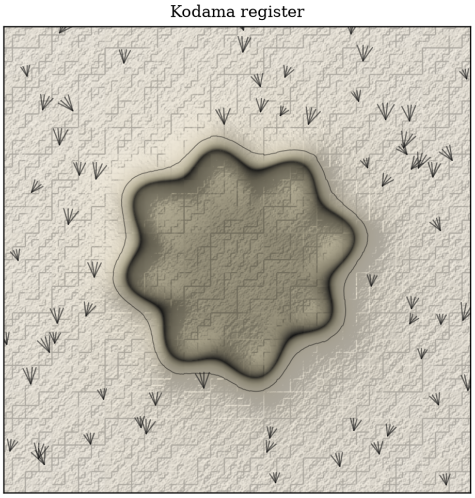

Figure 14. The Kodama register on its own. A single contour level and no vignette keep the field plain and evenly lit, matching the character's own unadorned, wooden design; field and palette are exactly those of Table 3 and Figure 18.

The No-Face register, named for the spirit Kaonashi ("No-Face") in *Spirited Away* (2001), is built around an absence rather than a feature. Napier's reading of the film treats its boundary-dissolving figures — Kaonashi foremost among them, a spirit with no fixed body of its own, only what it absorbs from others — as enactments of identity under threat of erasure (Napier 2006); the present register takes that absence literally, in field terms. Three metaballs are used, but only one is strong, sharply localized, and positive (the mask); the field is otherwise close to its own minimum across most of the silhouette, so that almost the entire body is, structurally, close to empty. The even harmonic order $k = 4$ gives a contour that no phase makes symmetric (Proposition 3.1): an identity the construction, like the film, declines to resolve into a fixed, symmetric shape. An anisotropy factor $\alpha = 0.72$ (Remark 1) elongates the silhouette vertically, toward a robed rather than a rounded shape.

No-Face is also the one register where the article's standard sampling grid is least kind to the texture it is meant to display: it has the lowest roughness exponent of the five ($H = 0.45$, Table 3), so its finest octaves carry more weight than any other register's (Figure 7) while being sampled on the same $N = 720$ grid as the rest. Figure 16 renders the identical field, parameters, palette, and presentation treatment as Figure

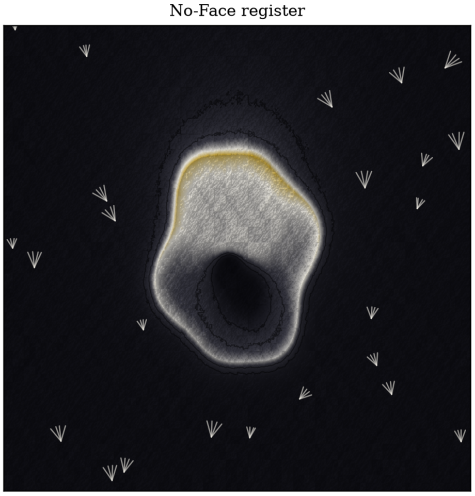

Figure 15. The No-Face register on its own. Two contour levels give the mask-like bright region and the dark hollow beneath it distinct edges; no vignette is applied. Field and palette are exactly those of Table 3 and Figure 18.

15, at three times the linear grid resolution ($N = 2160$), showing finer texture without the grid-scale moiré visible at $N = 720$.

The Catbus register, named for the grinning, twelve-legged Nekobasu of *My Neighbor Totoro* — the same 1988 film as the Totoro register, a different character from it — reuses Totoro's own low, gentle harmonic order $(3, 2)$, the same film's own visual grammar for warmth, but applies an anisotropy factor $\alpha = 1.55$ (Remark 1) to stretch the silhouette into a wide, bus-like body. Seven metaballs, rather than Totoro's five, stand in for a row of small legs along the underside and two bright headlight-eyes near the top; the roughness exponent ($H = 0.78$) is nearly as high as Totoro's own, for the same soft-fur reading. The result is deliberately the least uncanny of the five registers — no even-k guaranteed asymmetry, no non-crystallographic harmonic order, no near-empty field — consistent with the Catbus's own role in its film as an unambiguously friendly, helpful presence rather than an ambivalent or withheld one.

Figures 12–17 and 18 also carry a background layer of small tuft glyphs, scattered outside each register's own contour (1) . Their number is not a free per-figure choice: it is drawn from a Poisson process with intensity $\lambda = \lambda_{\max} H$, where H is that register's own wavelet roughness exponent from Table 3 — the same H already read, in Section 5, as a dial between coherent moss patches (H near 1) and unresolved, fur-like roughness (H near 0). Kodama and Totoro, already the two registers whose texture reads as mossiest, come out with the densest flora; No-Face, built around near-emptiness, comes out sparsest, without any register-specific tuning beyond the H each already has. This is, again, a presentation choice layered on top of Φ and the palette, not a change to either.

No-Face register, $N = 2160$ (detail)

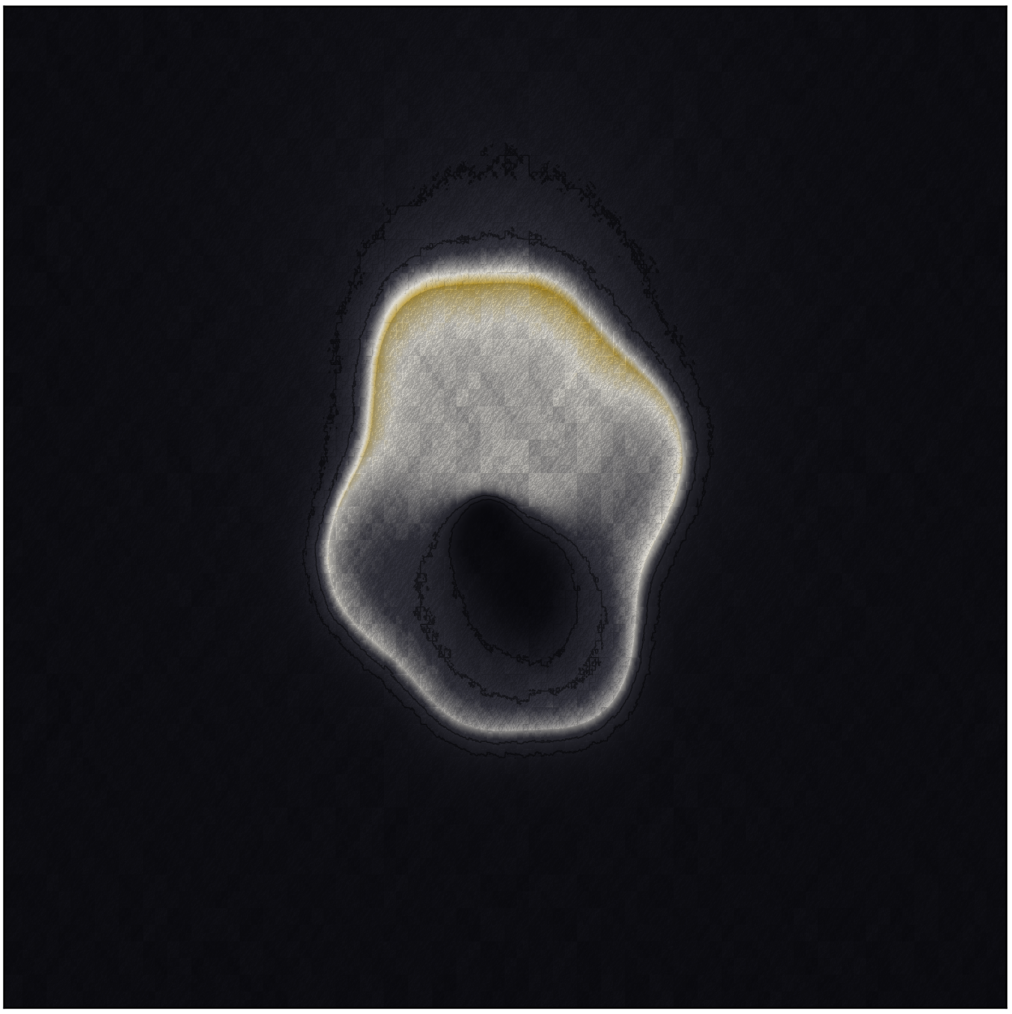

Figure 16. The No-Face register at $N = 2160$, three times the linear grid resolution of every other figure in this article. Field, parameters, palette, and contour treatment are unchanged from Figure 15; only the sampling grid is finer, which is why the mask's edge and the void beneath it read more finely grained here than elsewhere.

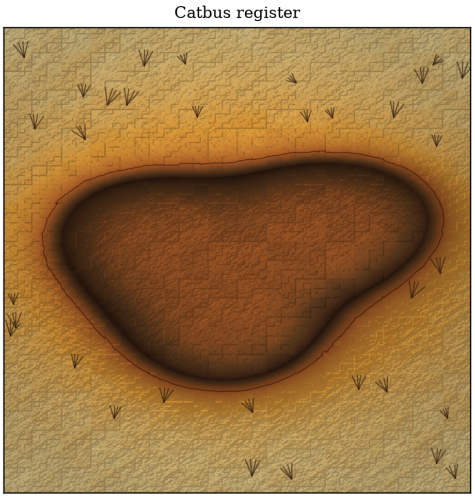

Figure 17. The Catbus register on its own. A warm vignette matches the treatment given to Totoro (Figure 12), the same film's other register; field and palette are exactly those of Table 3 and Figure 18.

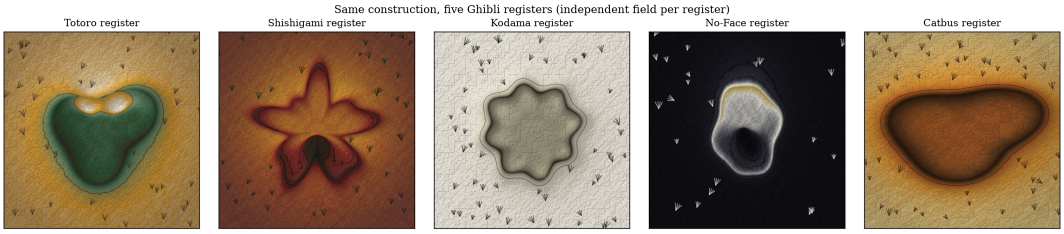

Figure 18. The same construction (Table 3), five Studio Ghibli registers, each an independent field and palette. Totoro (rounded, warm ink-moss-gold), Shishigami (star-lobed, night-green-to-gold), Kodama (pale, near-achromatic bone-to-ink), No-Face (void black with a mask-white and gold accent), and Catbus (warm amber-to-cream). Each panel also carries a register-specific *presentation* treatment layered on top of the same field and palette — a warm vignette for Totoro and Catbus, a paired day/night contour for Shishigami, a single flat contour for Kodama, two contour levels for No-Face, and a $\nabla\Phi$-based hillshade common to all five (Section 5.3) — disclosed here as cosmetic display choices, not changes to Φ or the color map itself. Enlarged individual renderings of all five registers appear as Figures 12, 13, 14, 15, and 17.

7. Quantifying the color grammar

Figure 19 gives the five palettes directly, as color swatches rather than as renders or as points in an abstract space: the raw material every number in this section is computed from. We refer back to it throughout.

Figure 19. The five palettes as sRGB swatches (exact hex values, the same values from which Table 4 and Figure 20 are computed). Shishigami's is the only four-color palette; the rest use five.

Color theory has long argued, informally, that warm hues (red, orange, gold) read as inviting and cool or narrow-range hues read as alarming or alien (Itten 1961). A far more rigorous source is now available: a 2025 systematic review of 132 peer-reviewed studies (42,266 participants, 64 countries) on color–emotion correspondences (Jonauskaite and Mohr 2025) reports that YELLOW and ORANGE are linked to positive, high-arousal affect, WHITE and light, low-chroma colors to positive but low-arousal affect, and near-achromatic BLACK to negative, high-arousal affect, GREY to negative, low-arousal affect specifically. Our five palettes let us test a sharper, quantitative version of these claims. Table 4 converts each palette's hex colors to CIELAB space (D65 white point, the same standard pipeline behind CIEDE2000 (Sharma, Wu, and Dalal 2005)) and reports, per palette, the mean lightness L^*, mean chroma $C^* = \sqrt{a^{*2} + b^{*2}}$, a chroma-weighted mean hue angle, a hue spread, and lightness range $L^*_{\max} - L^*_{\min}$ — a quantity needed to characterize No-Face's palette, whose reading is not well captured by mean lightness or chroma alone.

Hue at low chroma. Computing the hue angle $h = \text{atan2}(b^*, a^*)$ for every color of a palette, including colors with $C^* \approx 0$, and taking the maximum pairwise circular distance between all of them gives a misleading hue spread. At $C^* = 0$ the hue angle is undefined (Section 7.1); the number returned there is whatever `atan2` does with two rounding errors. Kodama's near-black (#1A1A1A) has $C^* = 0$, and over all five of Kodama's colors the hue spread would be 73.2°. The same problem distorts No-Face's mean hue: an unweighted circular mean of five hue angles, four of which belong to colors with $C^* < 8$, returns 321.9° (a blue-violet), although the only strongly chromatic color in the palette is a gold near 87°. Table 4 therefore reports (i) the hue spread over colors with $C^* \geq 5$ only, with the number of colors used, and (ii) a mean hue weighted by chroma, so that a color with no hue contributes no direction. The threshold 5 is a choice; Section 7.1 shows how much the statistic moves when it and the noise level are

varied.

Table 4. CIELAB summary of the five palettes (D65 white point), computed over the colors that have a hue (see below). Mean h is the circular mean of hue angles weighted by C^*; hue spread is the maximum pairwise circular distance among colors with $C^* \geq 5$ (number of such colors in parentheses); L^* range is $L^*_{\max} - L^*_{\min}$. The statistics over all colors are in Table 5.

Palette	mean L^*	mean C^*	mean h (deg) chroma-weighted	hue spread (deg) colours with $C^* \geq 5$	L^* range
Totoro	54.8	24.9	101.3	90.0 (4 of 5)	76.9
Shishigami	34.2	32.5	63.6	127.7 (4 of 4)	59.3
Kodama	65.5	8.5	95.4	7.5 (4 of 5)	85.7
No-Face	40.3	16.8	81.6	155.7 (3 of 5)	92.3
Catbus	50.4	37.7	69.4	50.2 (5 of 5)	80.6

Figure 20 shows the same data as points rather than averages: every color in every palette, plotted in the CIELAB a^*-b^* plane in its own true color. The two structural claims of this section are visible directly, not only in the summary statistics that follow. Catbus's crosses sit at large radius (high chroma) along a narrow angular wedge (narrow hue); Kodama's squares cluster near the origin (low chroma) in a tight column; No-Face's diamonds mostly cluster at or near the origin too, but one — the gold mask accent — sits far out on its own, the single most isolated point in the figure.

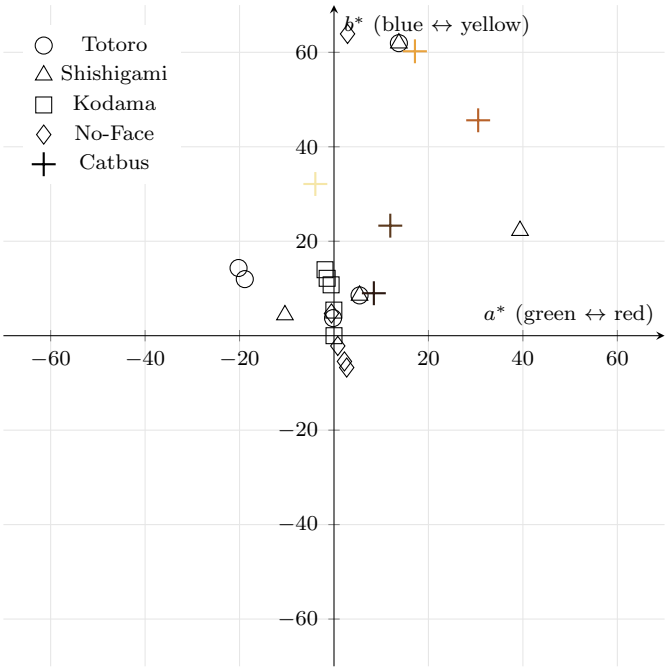

Figure 20. The five palettes as points in the CIELAB a^*-b^* (hue-chroma) plane, each marker drawn in its own true sRGB color (exact CIELAB coordinates). Radius from the origin is chroma; angle is hue. Catbus (crosses) is far from the origin in a narrow warm wedge; Kodama (squares) clusters tightly near the origin; No-Face (diamonds) mostly clusters at or near the origin, but its gold mask accent sits alone at large radius, the most isolated single point in the figure.

Chroma alone separates the two most extreme palettes cleanly. Catbus has the highest mean chroma of the five (37.7), all five of its colors are chromatic ($C^* \geq 5$), and their hue spread is 50.2°: a warm amber family running from pale cream to deep brown-red. It is chromatically the loudest register and also the one with no even-k asymmetry, no non-crystallographic harmonic, and the film's own friendliest reading

(Section 6). That combination places Catbus in the register Jonauskaite and Mohr (2025) associate with high-arousal positive affect for orange and yellow hues. The CIELAB numbers make the association more than a color-naming exercise: Catbus's hue angles cluster around 69° (chroma-weighted), squarely in the orange-gold band, while its lightness spans 80.6 L^* units rather than sitting at one saturated value, so "warm and friendly" here is carried by a hue band held nearly fixed while lightness and chroma vary: one axis held constant, the others left free to do the expressive work.

Kodama sits at the other chromatic extreme: its mean chroma (8.5) is the lowest of the five, under a quarter of Catbus's. Over the colors that have a hue, its four chromatic colors span only 7.5° of hue. Kodama is a single-hue palette (a yellowish tan, near 95°) that varies in lightness from near-white bone to one neutral near-black eye-dot; the earlier Table's 73.2° described the eye-dot's undefined angle, not the palette. That is consistent with the character's own canonical near-monochrome, pale-wood design (Section 6) and, broadly, with the positive, low-arousal reading Jonauskaite and Mohr (2025) report for light, low-chroma colors. It is also consistent with the film's treatment of kodama as a benign presence, despite a silhouette (Remark 2) drawn from the same forbidden harmonic-order set as the far more hostile Shishigami: the shape alone reads as unsettled, and the palette is doing real work to keep the overall register gentle rather than threatening. Among the five palettes, then, Catbus is the narrowest in hue only if one counts palettes in which every color has a hue; Kodama is narrower still over the colors it has.

No-Face resists both of the readings above. Its mean chroma (16.8) and hue spread (155.7°, still the widest of the five) look, from Table 4 alone, like a loud and varied palette; Figure 20 shows why that reading is misleading. Four of its five colors have $C^* < 8$ and the spread rests on three colors: two of them faint blue-greys ($C^* = 5.9$ and 7.3, near 292°) and the gold. What the statistic measures is a barely visible cool tint in the darks against one strongly warm accent, not many visibly different hues. What *is* robust is the one outlier: a single saturated gold point ($C^* \approx 64$) against four near-achromatic ones, and it is this structure — not mean hue, not mean chroma — that lightness range captures. No-Face's L^* range (92.3 units) is the widest of the five, spanning from the palette's darkest near-black ($L^* \approx 2.9$) to its near-white mask ($L^* \approx 95.2$), with the gold accent as a second, independent axis of contrast layered on top. The reading — vulnerable, faceless, absorbing whatever light or color reaches it except at one point that stays fixed — is carried by *range*, a variable the first version of this article introduced for exactly this palette.

7.1. *Hue is a polar coordinate, and it is singular at the origin*

The failure of the naive hue statistic has a mathematical cause, and the cause is a lesson. The map from (a^*, b^*) to (C^*, h) is the polar-coordinate map, with $dh = (a^*\,db^* - b^*\,da^*)/C^{*2}$. A perturbation of size δ in the a^*b^* plane therefore moves the hue by at most $\arcsin(\delta/C^*)$ when $C^* > \delta$, about δ/C^* radians for small δ, and by anything at all when $C^* \le \delta$: the same non-differentiability of θ at the origin that makes polar coordinates hard to learn (Moore, Paoletti, and Musgrave 2014). The middle panel of Figure 21 draws this as geometry: each color's uncertainty disc and the cone of hues it allows; the near-neutral colors of No-Face and Kodama get cones that are wide or the whole plane.

To see what this does to the palette-level statistic, we perturbed each color's (a^*, b^*) uniformly within a disc of radius $\delta = 2$, recomputed the *original* hue spread 4,000 times

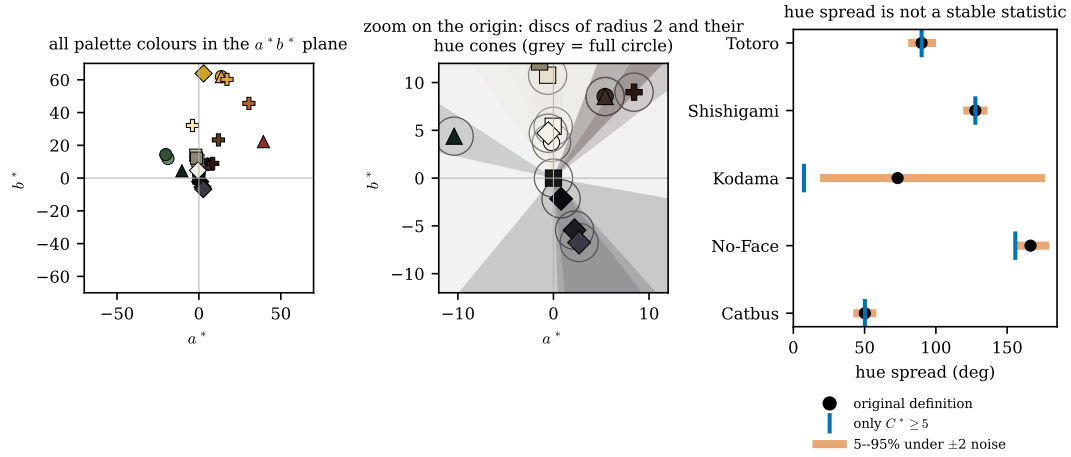

Figure 21. Hue is a polar coordinate. Left: every color of the five palettes in the CIELAB a^*b^* plane. Middle: zoom on the origin; each color has a disc of radius $\delta = 2$ (a perturbation of its (a^*, b^*)) and the cone of hue angles that disc allows (grey: the whole circle, when the disc contains the origin). Right: the palette-level hue spread under the original definition (black dot), with its 5–95% range under ±2 noise (vermillion bar, 4,000 draws) and the statistic over colors with $C^* \geq 5$ (blue tick). Marker shapes as in Figure 20.

per palette, and report its 5–95% range (Table 5). Only Kodama's statistic is unstable: its range is 157.9° wide (160.2° at $\delta = 1$, 148.7° at $\delta = 4$), because one of its colors has no hue. The other four ranges are narrow and grow with δ: Catbus's is 16.2° (8.1° at $\delta = 1$, 28.3° at $\delta = 4$), Totoro's 19.6°, Shishigami's 17.0°, No-Face's 23.2°. The noise radius is a choice, not a measured just-noticeable difference; we report three values so that the reader can see that the qualitative conclusion (one unstable palette, four stable ones) does not depend on it. The rule that follows is short: a hue statistic should be computed only over colors whose chroma is comfortably larger than the noise one cares about, and a mean hue should be weighted by chroma.

Table 5. Robustness of the hue spread. "orig." is the previous definition (all colors); the next two columns keep only colors with $C^* \geq 5$ and $C^* \geq 10$ ("–": fewer than two colors remain). "MC" is the median and 5–95% range of the original statistic when each color's (a^*, b^*) is perturbed within a disc of radius $\delta = 2$ (4,000 draws, fixed seed). The last column is the largest first-order hue shift $1/C^*$ (in degrees, capped at 180°) that a unit perturbation can cause in any color of the palette.

Palette	orig. spread	spread, $C^* \geq 5$	spread, $C^* \geq 10$	MC median (5–95%)	colours $C^* < 5$	max Δh per unit
Totoro	90.0	90.0	90.0	90.4 [80.5, 100.1]	1	15°
Shishigami	127.7	127.7	127.7	127.6 [119.2, 136.2]	0	6°
Kodama	73.2	7.5	4.5	96.9 [18.7, 176.6]	1	180°
No-Face	166.4	155.7	–	173.6 [156.3, 179.5]	2	25°
Catbus	50.2	50.2	50.2	50.1 [42.0, 58.2]	0	5°

In the classroom. Two short exercises use only what the section established. First, compute the Jacobian of $(a^*, b^*) \mapsto (C^*, h)$ and predict, before running anything, which palette's hue spread is fragile. Second, average the angles 350° and 10° arithmetically and by the resultant vector: the first gives 180°, the second 0°. A circular mean is necessary and, as the achromatic colors show, not sufficient.

7.2. *Legibility for readers with red–green color deficiency*

About one student in a class of thirty has red–green color deficiency (Section 2). For each palette we passed its colors through the Machado–Oliveira–Fernandes model of dichromacy (Machado, Oliveira, and Fernandes 2009), in the implementation of the `colour` library (protanopia and deuteranopia, severity 1.0; we do not simulate tritan deficiency because that implementation approximates it), converted to CIELAB, and computed the smallest CIEDE2000 difference ΔE_{00} between any two colors of the palette (Sharma, Wu, and Dalal 2005). Our implementation of ΔE_{00} reproduces test pair 1 of Sharma, Wu, and Dalal (2005) (2.0425) to four decimals. Table 6 gives the result. The Okabe–Ito triple used for the color-coded equation (Okabe and Ito 2008) is included as a reference.

Table 6. Smallest pairwise CIEDE2000 difference within each palette under normal vision and under simulated deuteranopia and protanopia (Machado et al. model, severity 1.0); "retained" is the deutan minimum divided by the normal one; the last column is the smallest lightness difference $|\Delta L^*|$ between any two colors under deuteranopia. The last row is the Okabe–Ito blue/vermillion/orange triple.

Palette	min ΔE_{00} normal	min ΔE_{00} deutan	min ΔE_{00} protan	retained (deutan)	min $\lvert\Delta L^*\rvert$ (deutan)
Totoro	22.8	8.7	12.2	38%	11.3
Shishigami	20.6	6.7	3.6	32%	7.6
Kodama	5.3	5.2	5.4	99%	5.5
No-Face	5.7	5.5	5.6	96%	7.7
Catbus	14.6	14.7	14.1	101%	17.7
Okabe–Ito triple	22.2	12.5	18.2	56%	14.2

Two groups appear. Catbus, Kodama and No-Face keep essentially all of their smallest contrast (101%, 99% and 96% retained under deuteranopia): Catbus and Kodama vary within one hue family, No-Face is mostly neutral, so lightness carries the contrast, and lightness is preserved. Totoro and Shishigami lose about two thirds: their limiting pair is a dark green against a dark brown (Totoro: 22.8 → 8.7 under deuteranopia) or a dark brown against a near-black green (Shishigami: 20.6 → 6.7, and 3.6 under protanopia), pairs that differ mainly along the red–green axis. Two cautions. The retention ratio is not the whole story: Kodama's and No-Face's smallest contrast is already only 5.3 and 5.7 under normal vision (adjacent bone tones; two near-blacks), so the table describes them as *sequential* maps that are fine as shading and poor as a set of category labels. And the reference triple, chosen for color-blind readers, retains only 56% of its contrast under deuteranopia (22.2 → 12.5), although that deutan minimum is still second in the table only to Catbus's (14.7); universal design is a matter of degree.

Figure 22 shows the consequence on one field. The bottom row keeps only L^*: in every panel the body is still separated from its surroundings, which is the design rule the table supports — let lightness carry structure and let hue carry mood.

In the classroom. Ask which of two palettes will survive deuteranopia, then check with the simulation. The mathematical content is a 3×3 matrix acting on linear RGB, a nonlinear gamma before it, and a metric (CIEDE2000) after it: a chain of maps whose composition students can inspect one link at a time.

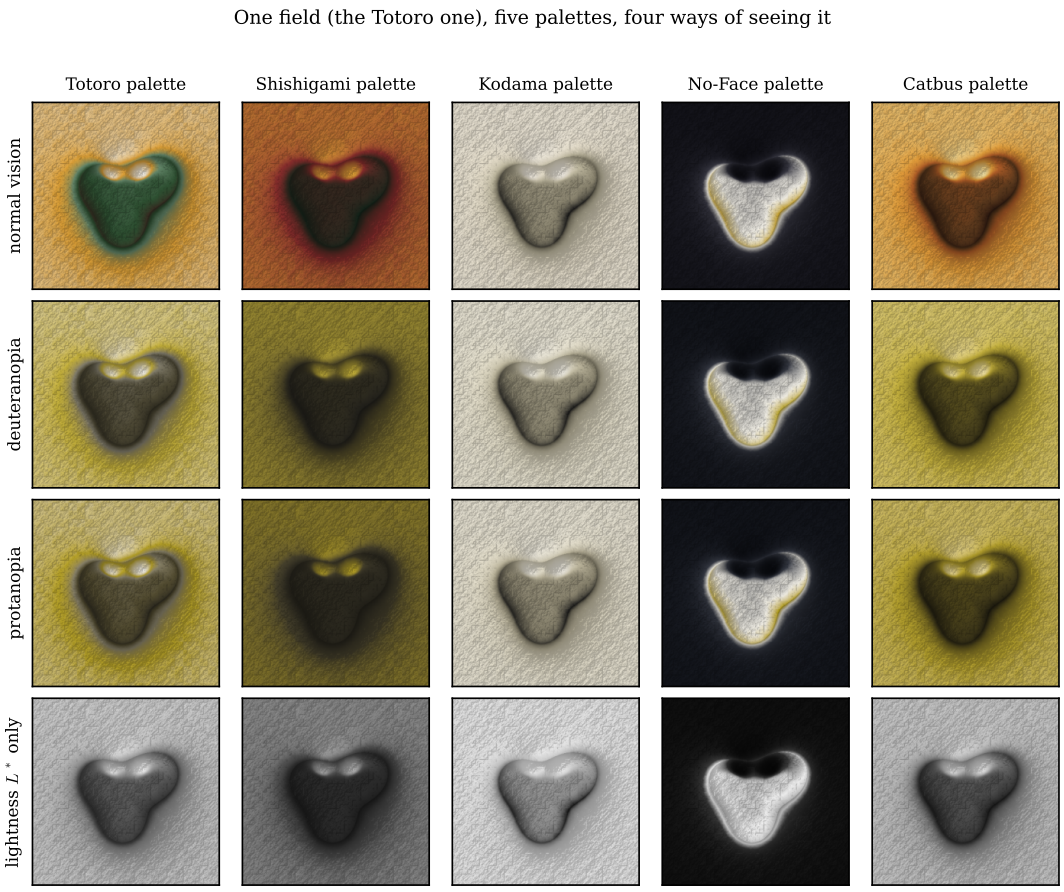

Figure 22. One field (the Totoro one), five palettes (columns), four ways of seeing it (rows): normal vision, simulated deuteranopia, simulated protanopia, and lightness L^* only. The simulation is that of Table 6.

7.3. *What the Pleasure–Arousal–Dominance model predicts for the five palettes*

The color–emotion literature gives one quantitative handle that needs only lightness and chroma: the standardized regressions of Valdez and Mehrabian (1994), $P = .69B + .22S$, $A = -.31B + .60S$, $D = -.76B + .32S$, for pleasure, arousal and dominance on brightness B and saturation S. We use each palette's mean L^* and mean C^*, standardized across the five palettes, as stand-ins for B and S. That substitution is ours and it is untested; only the *ranking* it produces is a prediction, and the units are not comparable to the original scale. Table 7 gives the values. Ranked by predicted arousal: Shishigami > Catbus > Totoro > No-Face > Kodama. By predicted pleasure: Kodama > Totoro > Catbus > No-Face > Shishigami. By predicted dominance: Shishigami > No-Face > Catbus > Totoro > Kodama.

Table 7. Predicted Pleasure, Arousal and Dominance ($\hat{P}, \hat{A}, \hat{D}$, standardized units) from mean L^* and mean C^* by the Valdez and Mehrabian (1994) regressions, as $\hat{P} = .69z_L + .22z_C$, $\hat{A} = -.31z_L + .60z_C$, $\hat{D} = -.76z_L + .32z_C$, with z the z-scores of the five palette means. A prediction to be tested (Section 8), not a result.

Palette	mean L^*	mean C^*	$\hat{P}$	$\hat{A}$	$\hat{D}$
Totoro	54.8	24.9	+0.38	-0.12	-0.38
Shishigami	34.2	32.5	-0.76	+0.90	+1.29
Kodama	65.5	8.5	+0.71	-1.35	-1.61
No-Face	40.3	16.8	-0.70	-0.17	+0.38
Catbus	50.4	37.7	+0.37	+0.74	+0.32

Where the prediction agrees with the qualitative literature it is reassuring, and where it does not it is informative. Kodama comes out highest in pleasure and lowest in arousal, matching the low-arousal, positive reading Jonauskaite and Mohr (2025) report for light, low-chroma colors; Catbus is high in arousal, matching their high-arousal reading of orange and yellow. No-Face is the disagreement. The same review links near-achromatic black to negative, high-arousal affect, whereas a mean-based prediction sees a dark palette of low chroma and puts No-Face in the lower half of the arousal ranking. The reason is structural: the mean of five colors cannot see that one of them is a saturated gold against blacks. This is the article's own L^*-range argument again, from the other side, and it is the place where a classroom test could most usefully disagree with the model.

7.4. *A computational check: shape versus color separability*

The CIELAB analysis above is a claim about *palettes*. A different, more basic question is whether the five *fields* Φ themselves — before any color map is applied at all — are already distinguishable, or whether it is the palette doing that work too. We tested this directly: two small convolutional networks were trained to classify a register from a single rendered sample, one network seeing only the grayscale field Φ (normalized to $[0, 1]$, no palette), the other seeing the same fields through their real palettes. Both were trained on 800 independent samples (five registers $\times$ 160 independent seeds each, at reduced grid resolution $N = 48$ for training speed) with an 80/20 train/test split.

Both classifiers reach 100% held-out test accuracy, on every register. This is an honest result, not a null one, but it sharpens rather than undermines the article's claim. Every register in Table 3 has its own (k, m) and its own metaball configuration: the five fields are geometrically distinct constructions, not one field wearing five palettes, so a classifier finding them shape-separable is confirming Table 3's own design, not

contradicting Section 7's. What the CIELAB analysis and Jonauskaite & Mohr's literature address is a narrower and, for this article, more important question than mere discriminability: not whether a register can be told apart from the others, but which specific *emotional register* — gentle, ambivalent, uncanny — a human viewer assigns to an already-distinguishable shape, a judgment a shape alone does not supply and this classifier was not asked to make.

7.5. *External validation against real film palettes*

Every CIELAB number so far has been computed from palettes designed for this construction. An independent check is available: the `ghibli` R package (Henderson 2024) distributes three tones (Light, Medium, Dark; seven colors each) per film, themselves derived from real film-still extractions by the Movies in Color project (Radulescu 2013) — external, published palettes for *My Neighbor Totoro*, *Princess Mononoke*, and *Spirited Away*, none authored by us. We converted all nine to CIELAB and compared each against its corresponding register(s): Totoro and Catbus against the Totoro palettes, Shishigami and Kodama against the Mononoke palettes, No-Face against the Spirited palettes.

One structural contrast is immediate and, on reflection, expected rather than a discrepancy: all nine real palettes have a hue spread between 167.7° and 179.4° (computed, as in Table 4, over colors with $C^* \geq 5$; over all colors it is 172.1° to 179.9°), close to the 180° maximum, because each is extracted from an entire film's worth of scenes and lighting, not one figure. Every register in Table 3 is narrower by design (the widest, No-Face's, is 155.7° over three colors; most are far narrower), because each isolates one character's reading rather than a film's whole cinematographic range. The real palettes are therefore not a target our registers should match in hue spread; they are a different quantity, a film's signature rather than a character's, and the gap between the two is itself evidence that the distinction in Section 3's own logic — one field, many readings — has a real-world analogue at the level of an entire film too.

Chroma is where the more informative agreement sits. Shishigami's mean chroma (32.5) sits within 4.6 of MononokeMedium's (37.1), the closest pairing of any register to any single real tone in this comparison; No-Face's (16.8) sits within 1.3 of SpiritedMedium's (18.2), closer still in absolute terms. Both were fixed independently of this check — Shishigami with the original field, No-Face for the lightness-range test of Section 7 — so neither number was tuned toward it. Kodama's chroma (8.5) sits nearest MononokeLight's (16.4), the real palette's own most desaturated tone, consistent with Kodama being designed paler than anything else in its film's register. Totoro and Catbus, both compared against the Totoro palettes, sit further from every real tone than the other three registers do from theirs (best combined $|\Delta L^*| + |\Delta C^*|$ around 28–37, against 12–16 for Shishigami, Kodama, and No-Face); we do not have a mechanism to offer for why the Totoro-film palettes are the harder match, only the honest observation that they are.

Remark 3. This is the article's central methodological point. "Gentle," "ambivalent," "near-achromatic," and "a single fixed accent against a void" are not being read off the images by appeal to taste; they are being connected to specific, measurable structure in CIELAB space (hue narrowness, chroma, and lightness range jointly) that is independently motivated by an external source, a 132-study psychological literature on color and emotion (Jonauskaite and Mohr 2025). The generative field in Section 3 supplies the shape; the palette supplies the reading; CIELAB space, read against

that literature and against this section's own honest caveat about hue instability at low chroma, is where that supply can be *related* to independent evidence — and, as Section 7.1 shows, where the relating can itself go wrong: a naive statistic (hue spread over all colors) misreads Kodama and overstates the case for No-Face. Relating is not testing; the test needs readers (Section 8.1).

8. The construction as a mathematics curriculum, and the claim that needs students

Table 8 lists what each step of the construction asks a student to do, and which figure or script lets them check it. The level column is our judgement of where the mathematics normally appears; no student has used this material, and Table 8 is a design argument, not an evaluation.

Table 8. Construction steps as curriculum topics. "Level" is the authors' estimate, not an observed grade. Every check in the third column appears in the cited figure or table.

Step	Mathematics	Level	Check a student can run
Contour $r(\theta)$	Polar coordinates; sum of sinusoids	upper secondary	Read the radial meaning of each term (Fig. 1)
Proposition 3.1	Parity of $\sin k\theta$; linear independence	first-year university	Predict which of twelve curves are symmetric (Fig. 2)
B, I fields	Exponential decay; logistic function	upper secondary	Change λ, β and watch the edge
Metaballs	Rational functions; critical points, Hessian, saddle	first-year calculus	Predict the merge threshold u_s (Fig. 3)
Excursion sets	Level sets; Euler characteristic $\chi = b_0 - b_1$	undergraduate	Count $\chi(K_u)$ against u (Fig. 9)
Proposition 5.1	Implicit function theorem; gradient as normal; first-order error	second-year	Halve Δu and see the error halve (Fig. 9)
Texture T	Step functions; geometric series 2^{-jH}; dyadic scales	upper secondary to first year	Octave variance shares (Table 2, Fig. 5)
Hue	Polar map on the color plane; Jacobian; circular mean	first-year university	Predict which palette's hue spread is fragile (Table 5)
Legibility	Matrix on linear RGB; gamma; a metric	first-year linear algebra	Simulate deuteranopia (Fig. 22)
Affect ranking	z-scores; a linear model; rank prediction	upper secondary statistics	Recompute Table 7

The figures of Sections 3 and 5 follow the tradition of Papert and of Abelson and diSessa, in which a learner changes one number and watches a picture change (Papert 1980; Abelson and diSessa 1981); the colored decomposition of Figure 1 follows Byrne (Byrne 1847) in letting color, not a letter, name the part. In Sinclair's terms the pictures do motivational, evaluative and generative work (Sinclair 2004): they invite a student in, they let the student judge that a curve is *wrong* (the mirror does not fit), and they suggest the next parameter to change.

8.1. *The one claim that needs students*

Every result above is computed from the palettes and the field. None is measured on a reader. The article's central claim, that with the shape and the task fixed the palette supplies the reading, is therefore supported only by the literature of Section 2, and by

a ranking (Table 7) that rests on our own substitution of mean L^* and mean C^* for the variables of Valdez and Mehrabian (1994). We state it as a prediction that a class could refute.

Design. Two versions of one worksheet, identical in mathematics, questions and layout (level-set questions on the field of Figure 9), differing only in palette: Catbus against Kodama. These two are chosen because they have the largest predicted arousal contrast among the palettes that also keep their internal contrast under deuteranopia (Table 6: 101% and 99% retained), so the contrast between arms is one of lightness and chroma and not of legibility. Students are randomized to a version within class. After the worksheet they rate their arousal and their valence on nine-point scales and answer the level-set questions.

Predictions, and what would refute them. *P1 (affect).* Mean arousal is higher with the Catbus palette than with the Kodama palette by at least 0.3 pooled standard deviations. P1 is refuted if the upper limit of the 95% confidence interval of the standardized difference is below 0.3. Detecting $d = 0.3$ at 80% power, $\alpha = .05$ two-sided, needs 176 students per arm (exact noncentral t). *P2 (performance).* Accuracy on the level-set questions differs between arms by less than 0.3 standard deviations in either direction, so that a teacher can choose a palette for its affect without paying for it in comprehension. P2 is refuted if the 95% interval lies wholly outside $[-0.3, 0.3]$. Showing equivalence within those bounds (two one-sided tests, true difference 0) needs 191 per arm, so about 2×191 students in all.

What each outcome would mean. If P1 fails, the palette does not supply the reading for these two palettes, the article's central claim lapses, and Table 7 is wrong at its widest gap. If P2 fails, the palette is not a free choice, and the finding of Plass et al. (2014) that color changes comprehension gains support in a mathematical setting. Either result is informative; a null on P1 would not be, unless the arousal rating is shown to be sensitive to some other manipulation.

Where the design is weak. The two palettes differ in L^* and C^* together, so an effect cannot be assigned to lightness or to chroma separately; that needs a 2×2 design and at least four arms. A rating of a worksheet is not a rating of a creature. And the classroom setting invites demand effects that a laboratory does not.

9. Discussion

The field Φ of Equation (6) is aesthetically inert. Its harmonics, its metaball topology, and its wavelet roughness exponent determine a shape; nothing in the equations refers to a forest spirit, a tree, or a mask. The reading is supplied entirely by the color map, and Section 7 relates that supply to measurable quantities: chroma, lightness, lightness range, and (with care) hue, in CIELAB space, quantities with an independent empirical history — a 132-study literature on color and emotion (Jonauskaite and Mohr 2025) — that predates wavelets, metaballs, and Miyazaki altogether. Whether those quantities predict what a *reader* sees is a claim about readers, and this article has none; Section 8.1 says how it could be refuted.

The No-Face register makes a point the other four cannot make on their own, because *Spirited Away* withholds, structurally, exactly what the other four films supply. Kaonashi has no face of its own, no fixed voice, and no body beyond what it absorbs from what it consumes; Napier's reading treats this as a figure of identity under threat

of erasure (Napier 2006), rather than as a fixed character design in the way Totoro or the Catbus are fixed designs. Applying the same generative field and the same CIELAB machinery to a subject whose own narrative function is to resist having a settled visual identity is not a contradiction of that function; it is a demonstration, at one remove, of the article's own thesis — that a shape and a color map are separable, and that the second, not the first, is where a viewer's unease or tenderness actually lives. Totoro and the Catbus supply shape and color already fused into a beloved, stable image; No-Face's own narrative arc supplies neither fixedly, on purpose, shifting from menacing to sympathetic across the film; the present construction reflects that instability formally, in a field that is empty almost everywhere except at one point that never moves.

This suggests a modest general claim for teaching and for generative art practice: when a single parameterized field is meant to support multiple readings, the palette-design problem can be posed quantitatively, in a perceptually uniform color space, rather than negotiated purely by eye. We do not claim CIELAB chroma, hue spread and lightness range are the only variables that matter for emotional reading, only that these five palettes already need all three: chroma separates Catbus (highest) from Kodama (lowest); hue spread, computed over colors that have a hue, separates the single-hue palettes (Kodama, Catbus) from the multi-hue ones (Totoro, Shishigami, and No-Face with its faint cool tint against a gold); and lightness range is needed specifically to place No-Face, whose mean lightness and mean chroma alone would not distinguish it from several other near-achromatic palette segments. A systematic study, varying one CIELAB quantity at a time across a larger, purpose-built palette set, is left for future work.

The five parameters (a, b, k, m, ϕ) governing the contour, together with the metaball centers, weights, wavelet exponents, and (for Catbus and No-Face) the anisotropy factor α, form a single vector that could equally be called a genotype, with the rendered image as its phenotype; a genetic algorithm could search this space under a fitness function scoring legibility or charm, mutating and recombining registers the way Section 6 constructed them by hand. We do not pursue that search here — it is a different paper, about optimization rather than about the color-shape separation this one argues for — but the vector itself, Table 3 read as five points in one parameter space rather than five unrelated designs, is already the object such a search would explore.

Every register in this article is built on the same domain: the polar plane (ρ, θ) of Section 3, an anisotropically stretched version of it for Catbus and No-Face (Remark 1), but a plane nonetheless. Nothing in Eq. (1)–(6) is specific to that choice of domain: ρ and θ could equally be geodesic distance and azimuth on a sphere, or the two periodic coordinates of a torus, with the same contour, boundary, interior, metaball, and wavelet-texture construction carried over verbatim once "distance" and "angle" are reinterpreted in the new geometry. A spherical register would read, quite differently, as a small planet rather than a bounded creature; a toroidal one would force the wavelet texture to tile without a seam, a constraint the present planar construction never has to meet. We do not build either register here — it changes what the excursion set K_u of Eq. (7) means topologically, which is a genuine change of subject, not a parameter choice — but flag it as the most direct way this construction could leave the plane.

Two disclosures and one open claim. First, the palettes are not equally independent of the measurement: Totoro, Shishigami and Catbus's amber family were fixed toward each character's canonical, documented design before any CIELAB computation, whereas Kodama's near-achromatic palette and, especially, No-Face's lightness-

spanning one were designed with the measurement already in mind — No-Face's to test whether a single fixed accent against a near-empty field would register as a distinct CIELAB signature. Those two are targeted constructions, not independent data points that happened to confirm the hypothesis. Second, each palette has four or five colors, so every mean and range rests on few points: No-Face's lightness range on two colors, its hue spread on three; Section 7.1 quantifies how fragile the hue statistic is and Table 4 reports how many colors each number rests on. The open claim is the one that matters, and it is the one Section 8.1 states as a prediction with a refutation criterion: that the palette, and not the shape, supplies the reading a student sees.

10. Conclusion

A generative field is a good object for teaching two things at once, and the two can be kept apart. What a class can *check* it has checked: the symmetry proposition in 2,952 of 2,952 cases, the threshold-flow law with error proportional to the step, the metaball merge threshold to 0.32%. What it can only *predict*, it has predicted, with a sample size and a criterion for being wrong. For a teacher choosing a palette for a mathematical figure the article supports five rules.

(1) Let lightness carry structure and hue carry mood; then check the palette under simulated deuteranopia and protanopia (Table 6). A palette in one hue family survives; a pair of dark green and dark brown does not (Totoro and Shishigami lose about two thirds of their smallest contrast).
(2) Compute hue statistics only over colors whose chroma is well above the noise one cares about, and weight mean hue by chroma: over all of Kodama's colors the hue spread is 73.2°, over the colors that have a hue it is 7.5°.
(3) Give each mathematical object one color and keep it across equation, figure and text (Ozcelik et al. 2009; Byrne 1847), and never let color alone carry the distinction.
(4) Do not build a design on the claim that red impairs performance (Gnambs 2020); build it on the associations that have replicated (Jonauskaite et al. 2020), and state them as associations.
(5) Treat the Pleasure–Arousal–Dominance ranking of Table 7 as a hypothesis until 176 students per arm have said otherwise (Section 8.1).

Acknowledgments

The generative field and the Totoro and Shishigami registers come from a teaching notebook for a general audience; the mathematics and the color analysis are developed here.

Disclosure statement

The author reports there are no competing interests to declare.

Data availability statement

The code that computes every figure, table and number of this article from Equation (6) and the five palettes is available from the author.

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Author biography

César Galindo is a mathematician in the Facultad de Empresariales, Universidad Panamericana, Ciudad de México. He works in algebraic geometry and homotopy theory, and in the geometry of generative images.